

Linear Models

Geometric Algorithms

Practice Problem

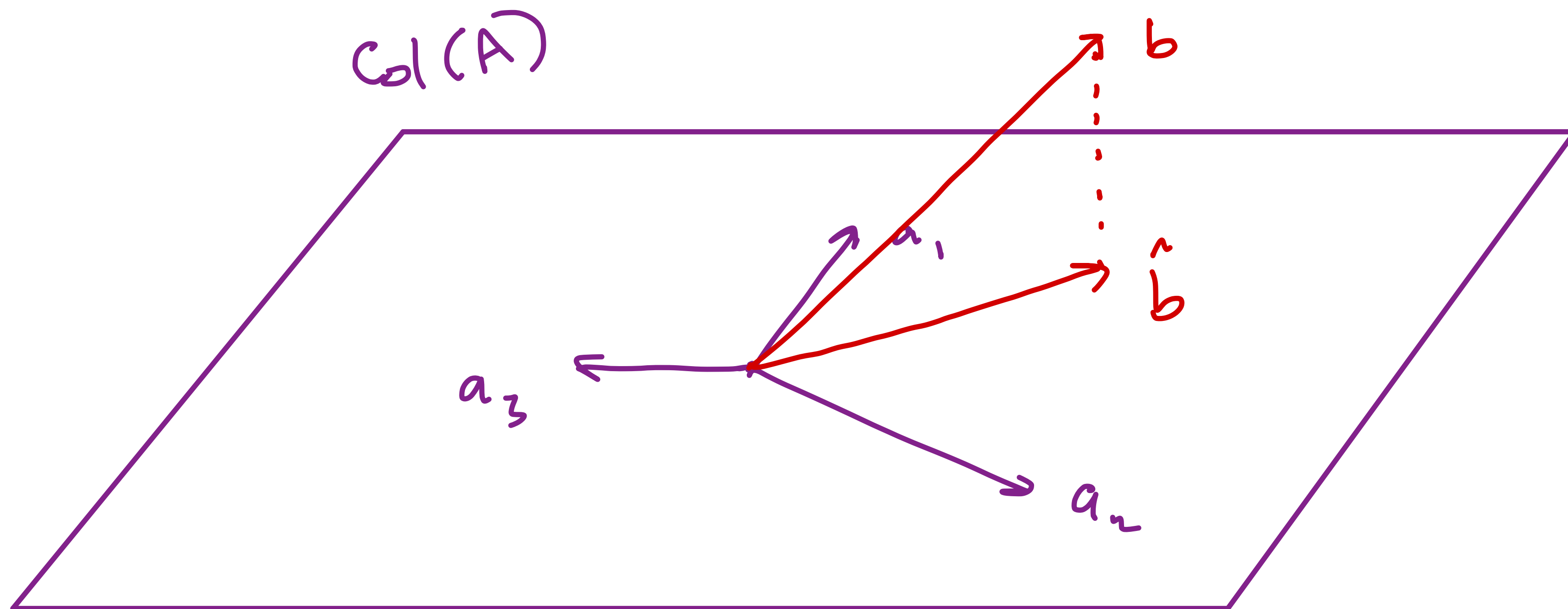
$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix}$$

Find the ^{orthogonal} projection of $\mathbf{b}$ onto $\text{Col}(A)$

(note that $\mathbf{a}_3 = \mathbf{a}_2 - \mathbf{a}_1$)

Answer

$$A = \begin{matrix} & \begin{matrix} a_1 & a_2 & a_3 \end{matrix} \\ \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix} & \end{matrix} \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix}$$



$$A^T A \hat{x} = A^T b$$

$$\text{Given } \hat{x}, \quad \vec{b} = A \hat{x}$$

Answer

$$a_3 \in \text{span}\{a_1, a_2\}$$

$$A = \begin{matrix} & a_1 & a_2 & a_3 \\ \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix} & & & \end{matrix} \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix}$$

$$\text{proj}_{\text{Col}(A)} \mathbf{b} = \text{proj}_{\{a_1, a_2\}} \mathbf{b}$$

$$a_1, a_2 \text{ are L.I.} \Rightarrow \{a_1, a_2\} \text{ is basis } \text{Col}(A)$$

$$X = [a_1 \ a_2]$$

$$X \vec{x} = \vec{b}$$

same solutions as

$$X \vec{x} = \hat{\vec{b}}$$

$$X^T X \vec{x} = X^T \vec{b} \text{ has unique sol.}$$

$$\vec{x} = (X^T X)^{-1} X^T \vec{b} \text{ and}$$

$$\hat{\vec{b}} = X (X^T X)^{-1} X^T \mathbf{b}$$

Answer

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{10 - 4} \begin{bmatrix} 5 & -2 \\ -2 & 2 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 5 & -2 \\ -2 & 2 \end{bmatrix}$$

$$X^T \mathbf{b} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix} = \begin{bmatrix} -1 \\ 7 \end{bmatrix}$$

$$\hat{\mathbf{b}} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \left(\frac{1}{6} \right) \begin{bmatrix} 5 & -2 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 7 \end{bmatrix}$$

Objectives

1. Use the least square method to build linear *models* of noisy data
2. Show how we can use linear algebraic methods to model with non-linear models

Keywords

line of best fit

independent/dependent variables

residuals

prediction

simple least squares regression

multiple regression

polynomial regression

models

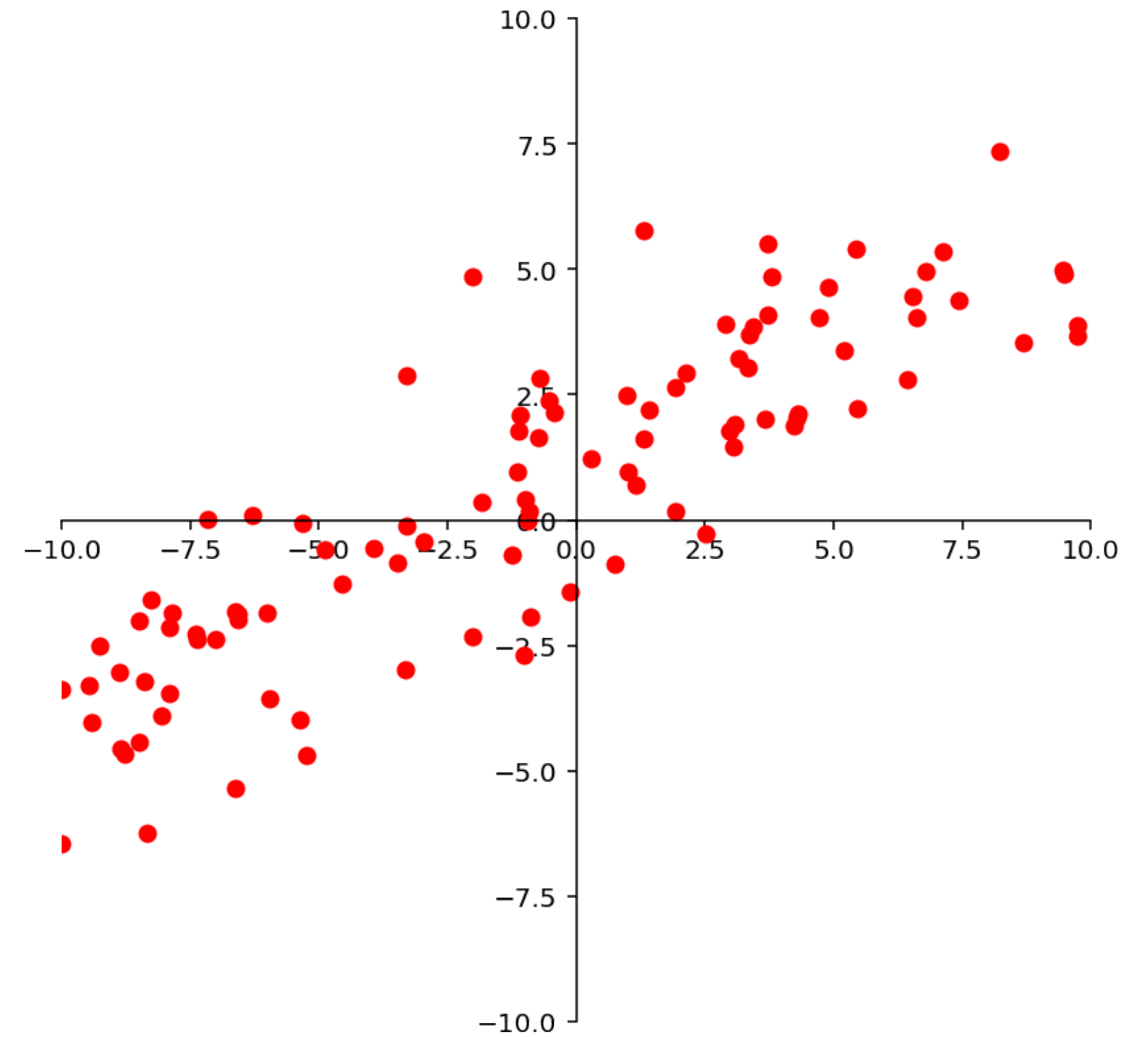
model fitting

model parameters

design matrices

Warm-up: Line of Best Fit

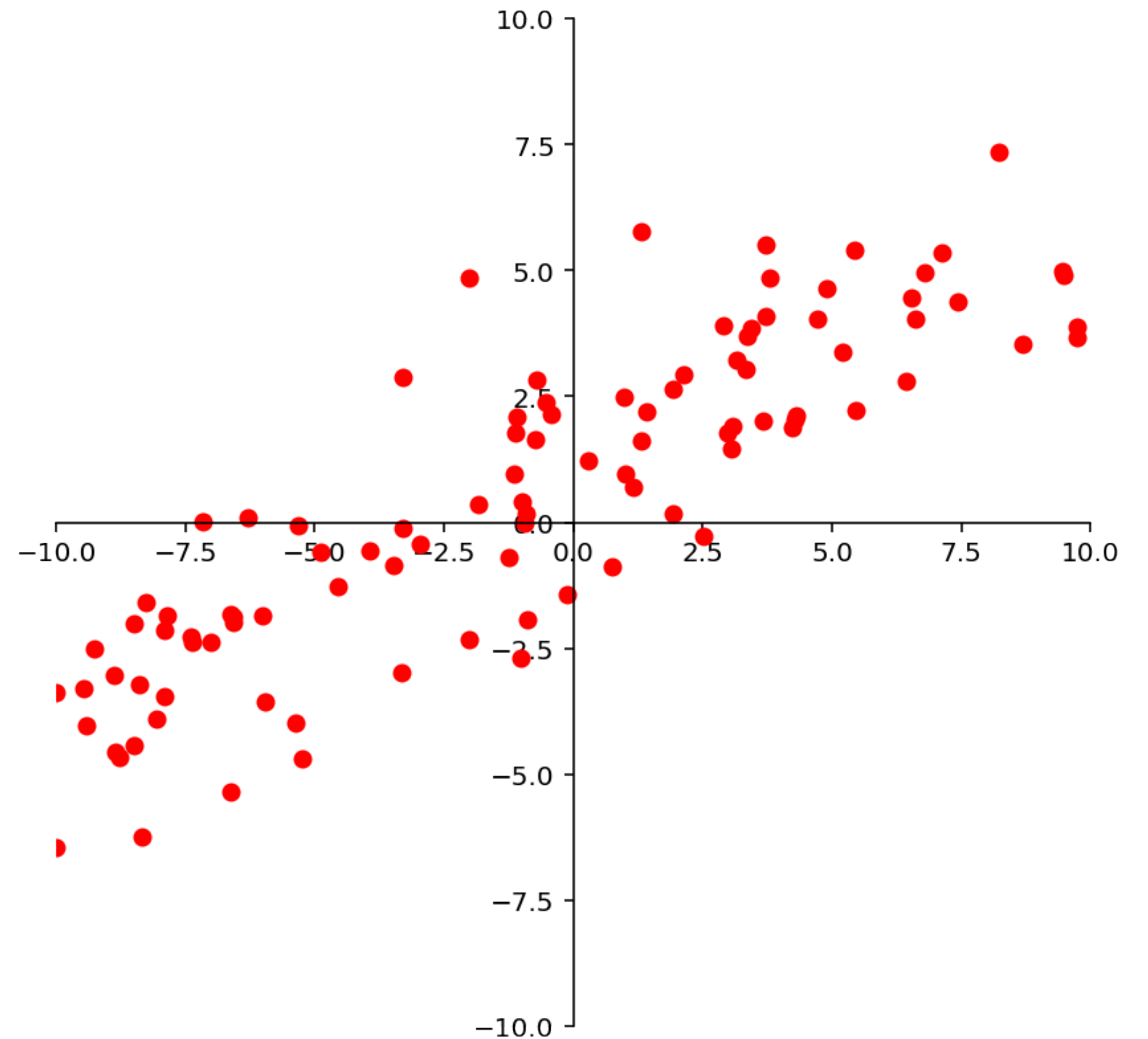
The Setup



The Setup

You're given points in $\mathbb{R}^2$:

$$\{(x_1, y_1), \dots, (x_k, y_k)\}$$

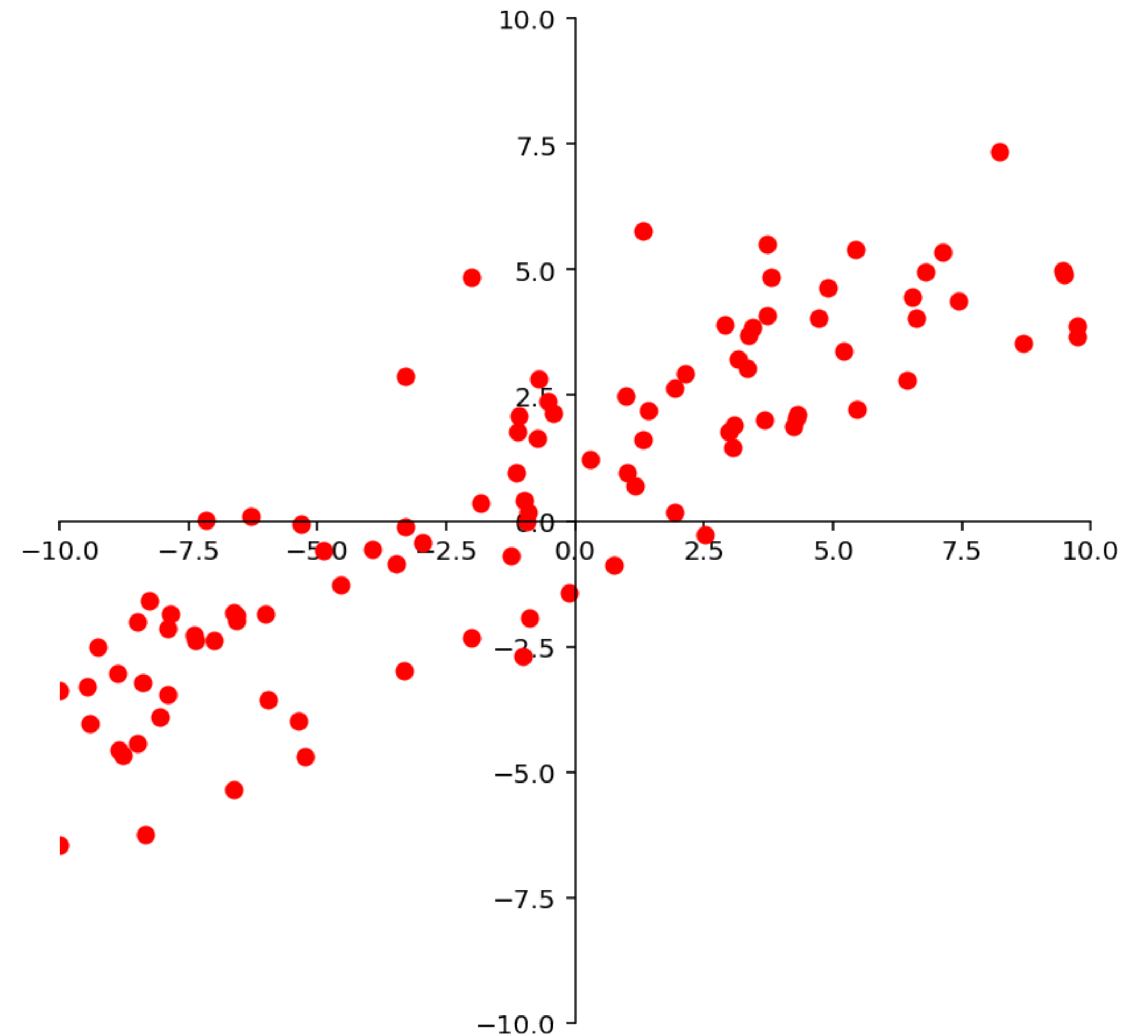


The Setup

You're given points in $\mathbb{R}^2$:

$$\{(x_1, y_1), \dots, (x_k, y_k)\}$$

Ex. (height, weight) data for a population



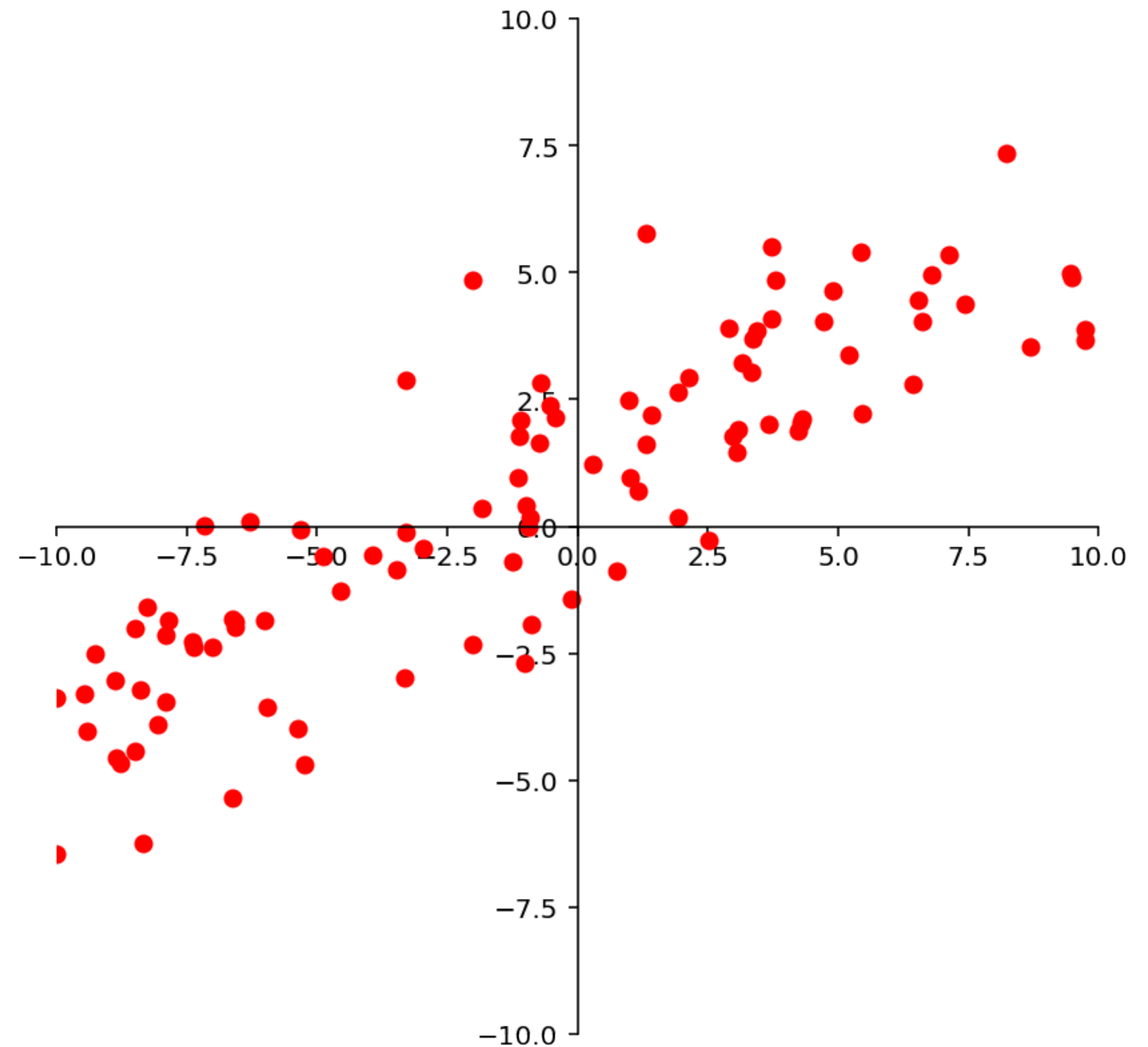
The Setup

You're given points in $\mathbb{R}^2$:

$$\{(x_1, y_1), \dots, (x_k, y_k)\}$$

Ex. (height, weight) data for a population

The data *kind of* trends
as a line



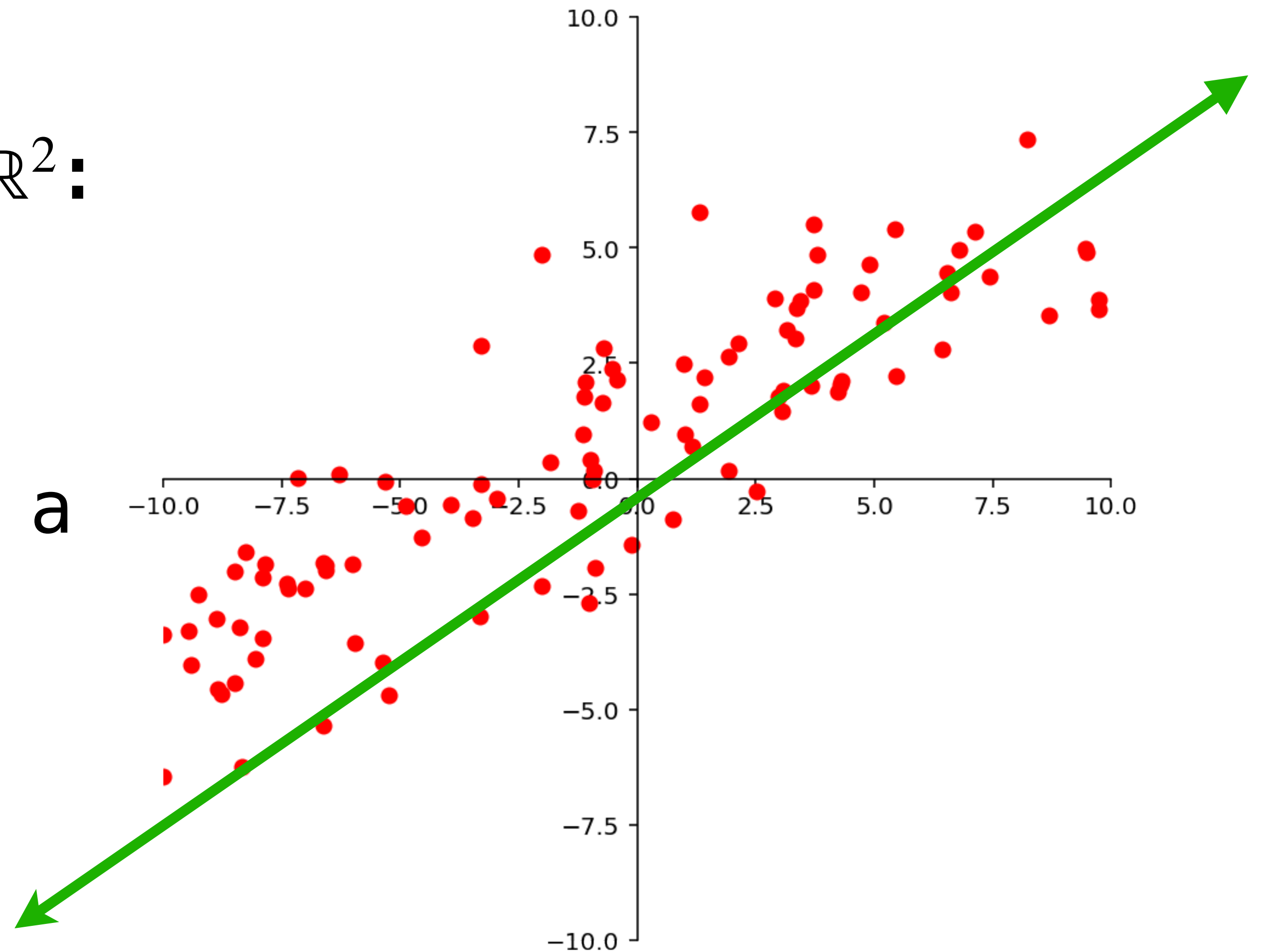
The Setup

You're given points in $\mathbb{R}^2$:

$$\{(x_1, y_1), \dots, (x_k, y_k)\}$$

Ex. (height, weight) data for a population

The data *kind of* trends as a line



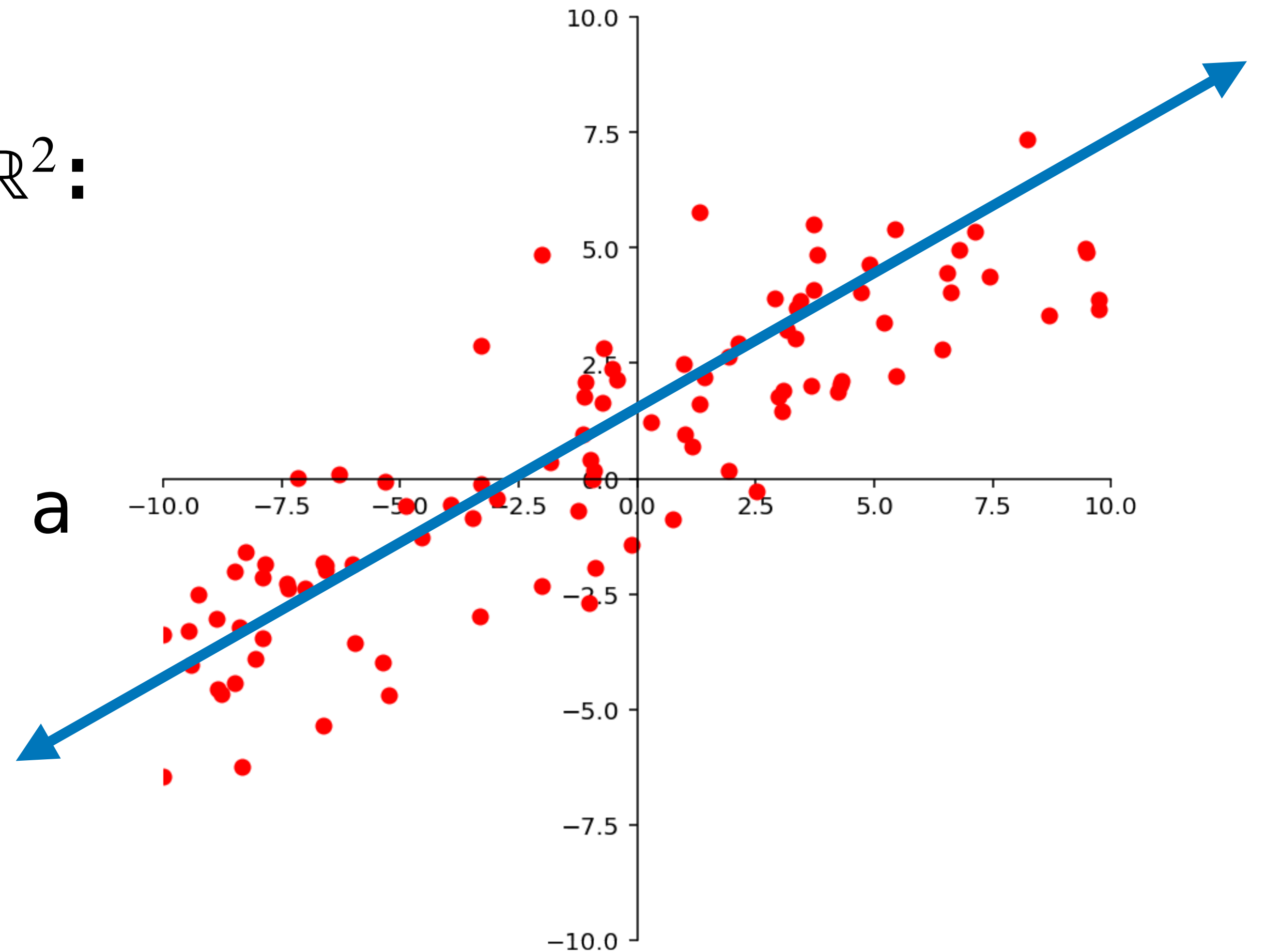
The Setup

You're given points in $\mathbb{R}^2$:

$$\{(x_1, y_1), \dots, (x_k, y_k)\}$$

Ex. (height, weight) data for a population

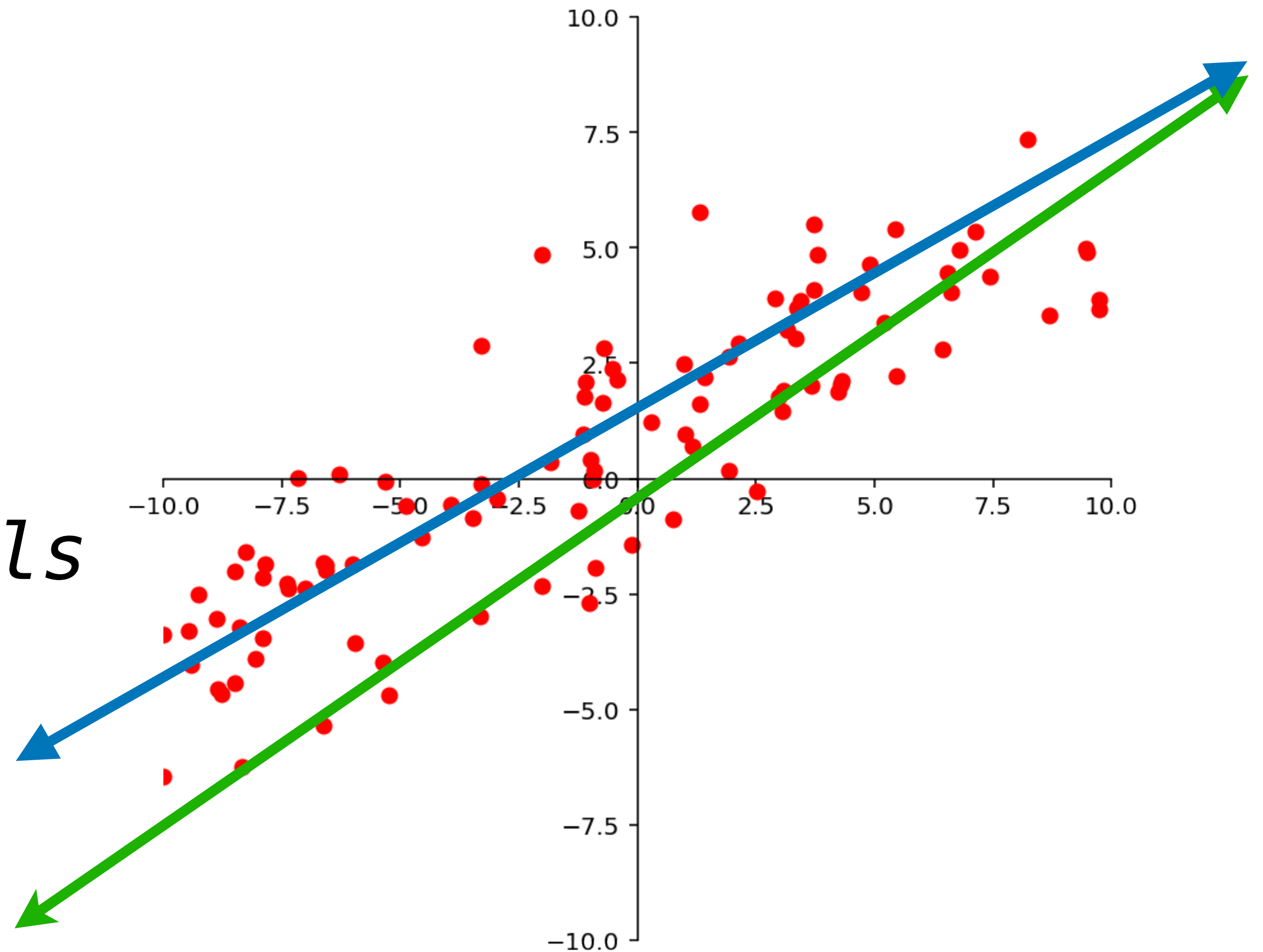
The data *kind of* trends as a line



The Setup

Q. Which line "best" describes the trend of the dataset?

Q. Which one best models the dataset?



Two Important Questions

Two Important Questions

1. What is a model?

Two Important Questions

1. What is a model?

We'll come back to this...

Two Important Questions

1. What is a model?

We'll come back to this...

2. What does "best" mean?

Two Important Questions

1. What is a model?

We'll come back to this...

2. What does "best" mean?

This is a make-or-break question

Least Squares Simple Linear Regression

Q. Given a set of points $\{(x_1, y_1), \dots, (x_n, y_n)\}$, find the line

$$f(x) = \beta_0 + \beta_1 x$$

which *minimizes*

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

Least Squares Simple Linear Regression

Q. Given a set of points $\{(x_1, y_1), \dots, (x_n, y_n)\}$, find the line

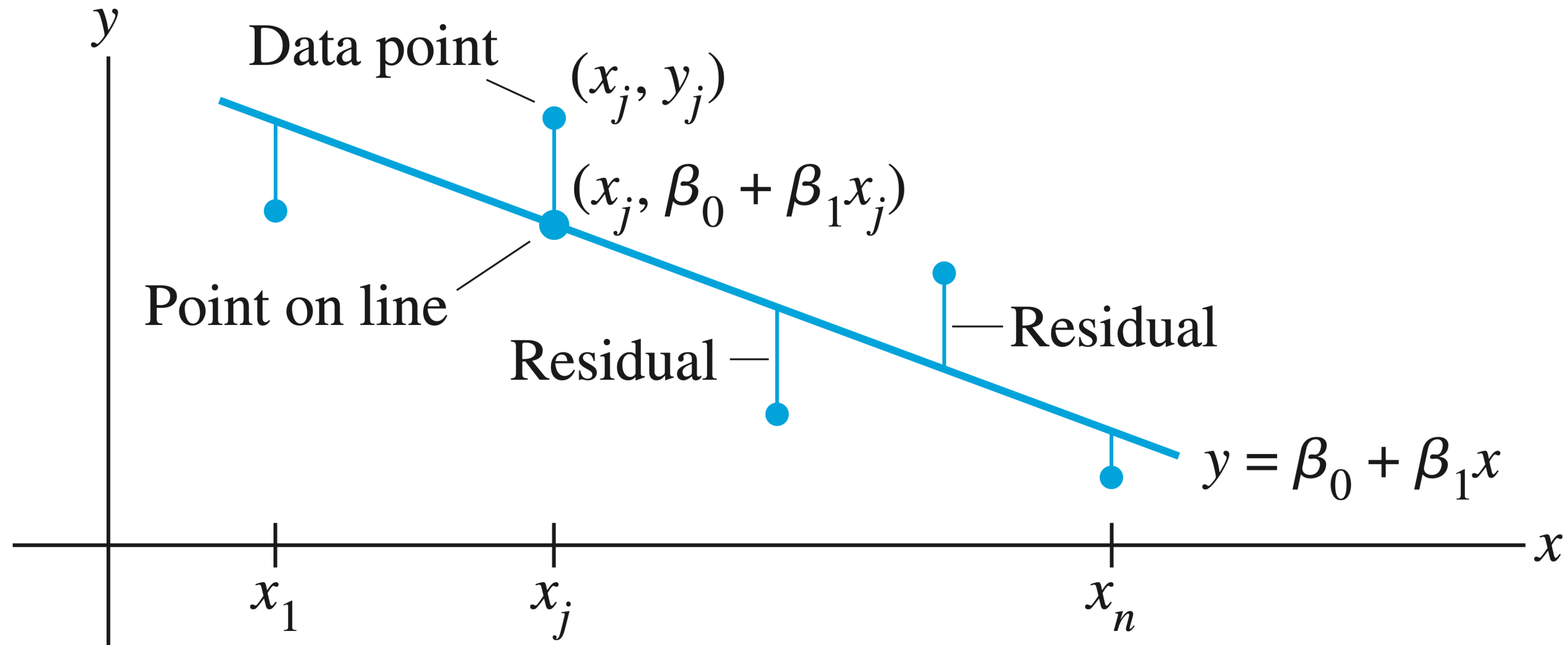
$$f(x) = \beta_0 + \beta_1 x$$

which *minimizes*

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

The "best" line
minimizes the *sum
of squares of
differences*

The Picture



Find the line which makes the (squared) sum of these differences *as small as possible*

An Aside: Is this really "best"?



An Aside: Is this really "best"?

Who's to say...



An Aside: Is this really "best"?

Who's to say...

Depends on the data, the
application domain, etc.



An Aside: Is this really "best"?

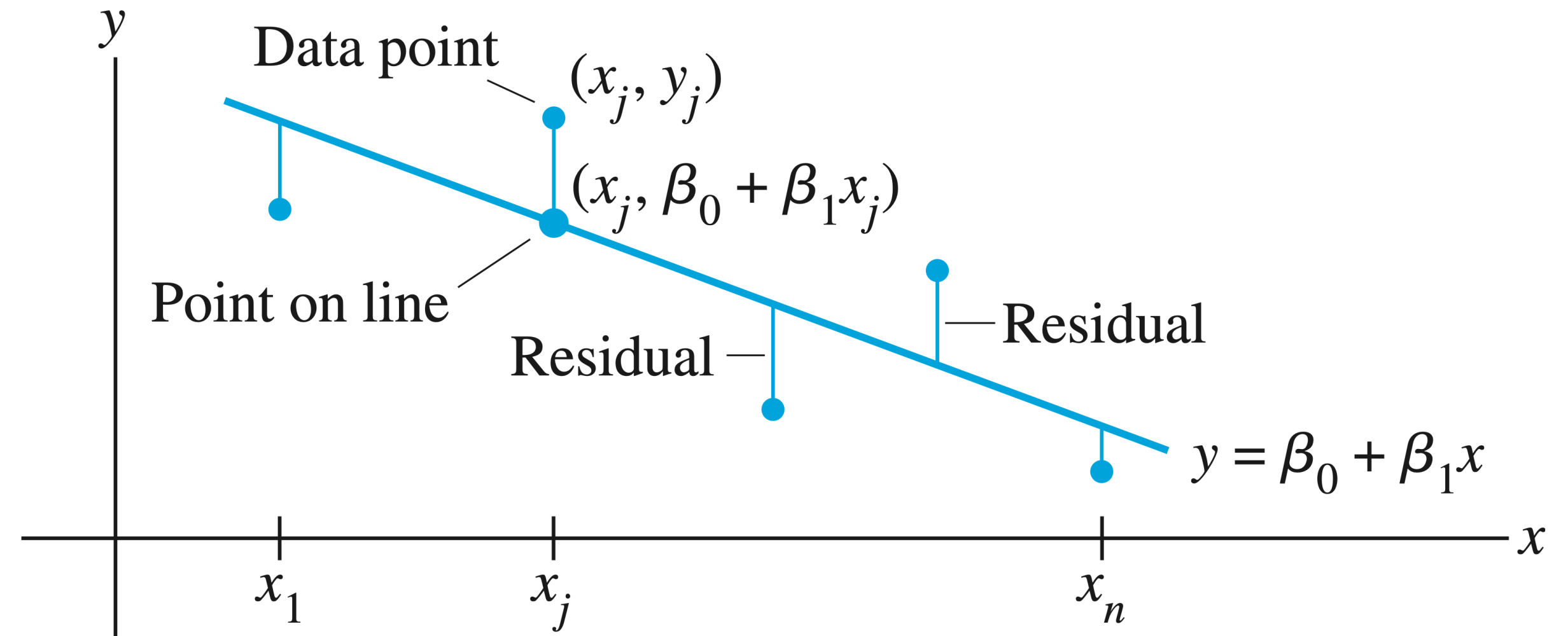
Who's to say...

Depends on the data, the application domain, etc.

The point. We fix the notion of "best" and then we do calculations



Terminology



$$\{(x_1, y_1), \dots, \overset{\text{data point}}{(x_i, y_i)}, \dots, (x_n, y_n)\}$$

dataset

$$f(x) = \overset{\text{model parameters/ regression coefficients}}{\beta_0} + \overset{\text{model}}{\beta_1 x}$$

independent variable

$$(x_i, y_i)$$

dependent variable (label)

$$\sum_{i=1}^n \overset{\text{observation}}{(y_i - \overset{\text{prediction}}{f(x_i)})^2}$$

residual

Terminology: Datasets

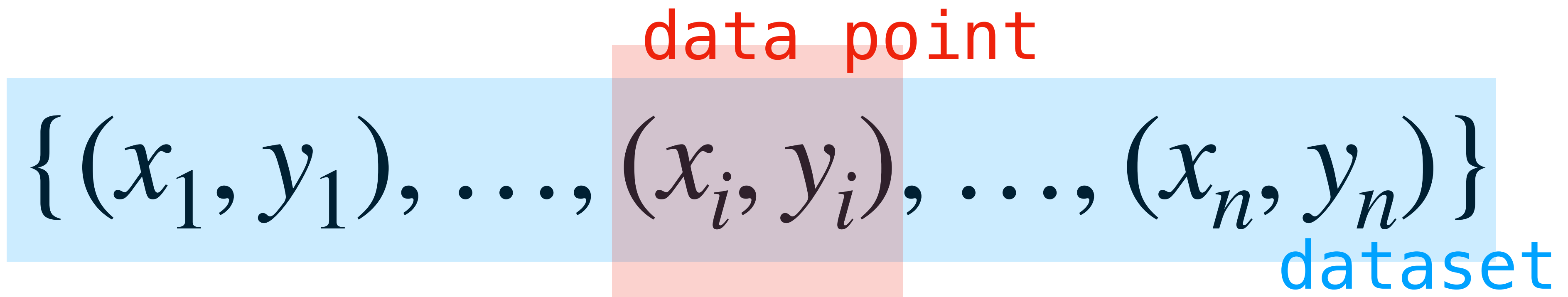
$$\{(x_1, y_1), \dots, (x_i, y_i), \dots, (x_n, y_n)\}$$

Terminology: Datasets

$$\{(x_1, y_1), \dots, (x_i, y_i), \dots, (x_n, y_n)\}$$

dataset

Terminology: Datasets



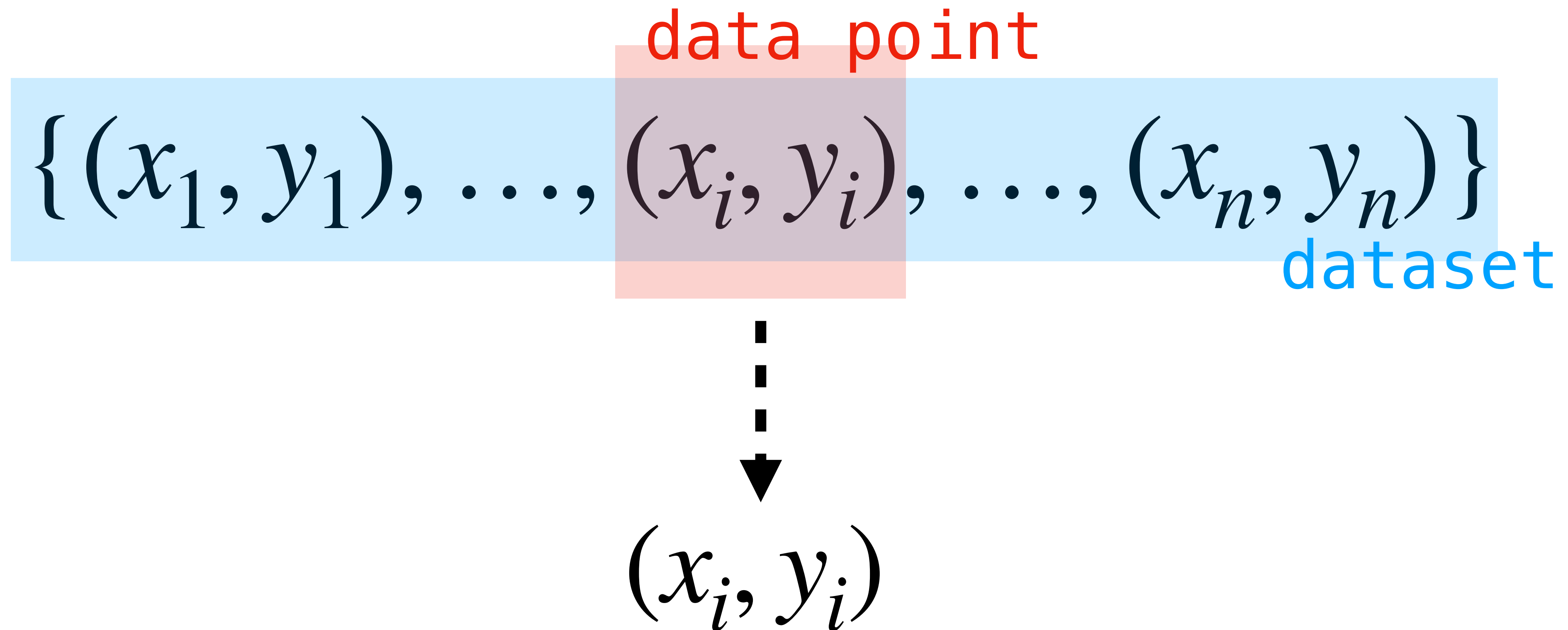
The diagram illustrates the terminology for datasets. It features a light blue rectangular background representing the entire dataset. Within this background, a smaller, semi-transparent light red rectangle highlights a specific data point. The text $\{(x_1, y_1), \dots, (x_i, y_i), \dots, (x_n, y_n)\}$ is centered within the blue area. The label "data point" in red is positioned above the highlighted red rectangle, and the label "dataset" in blue is positioned below the right side of the blue rectangle.

$$\{(x_1, y_1), \dots, (x_i, y_i), \dots, (x_n, y_n)\}$$

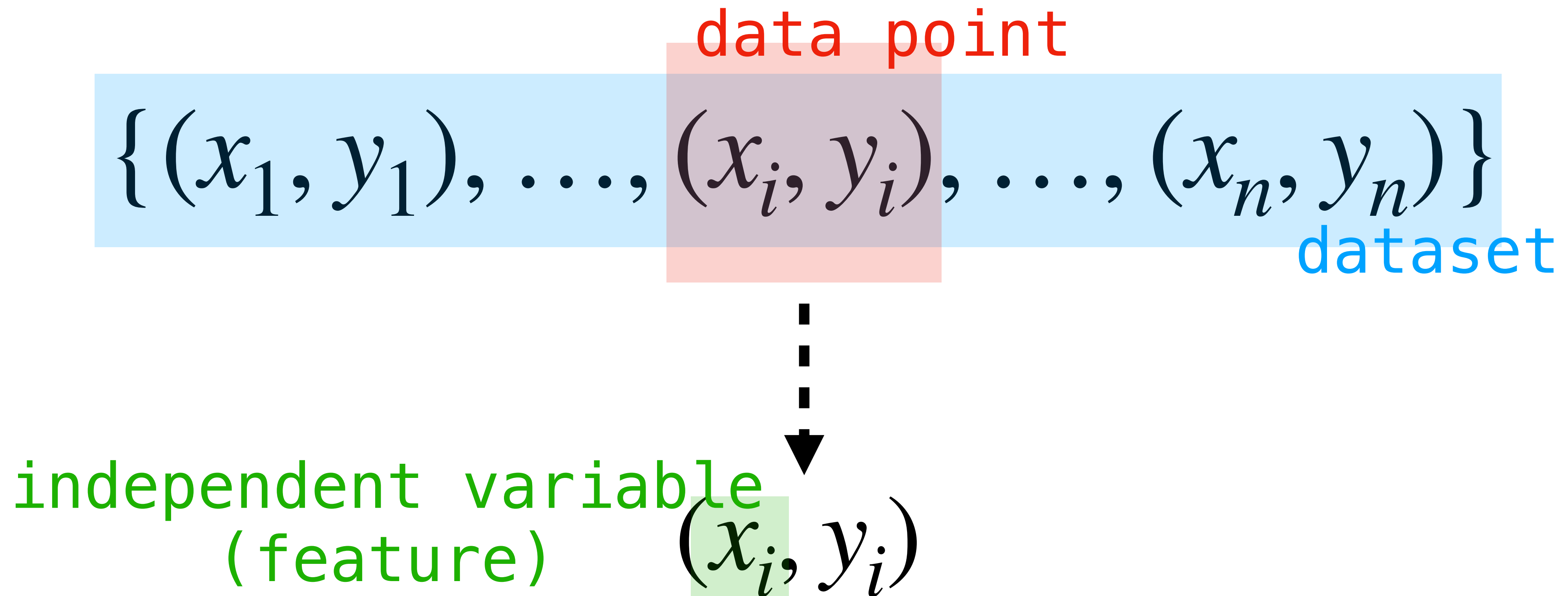
data point

dataset

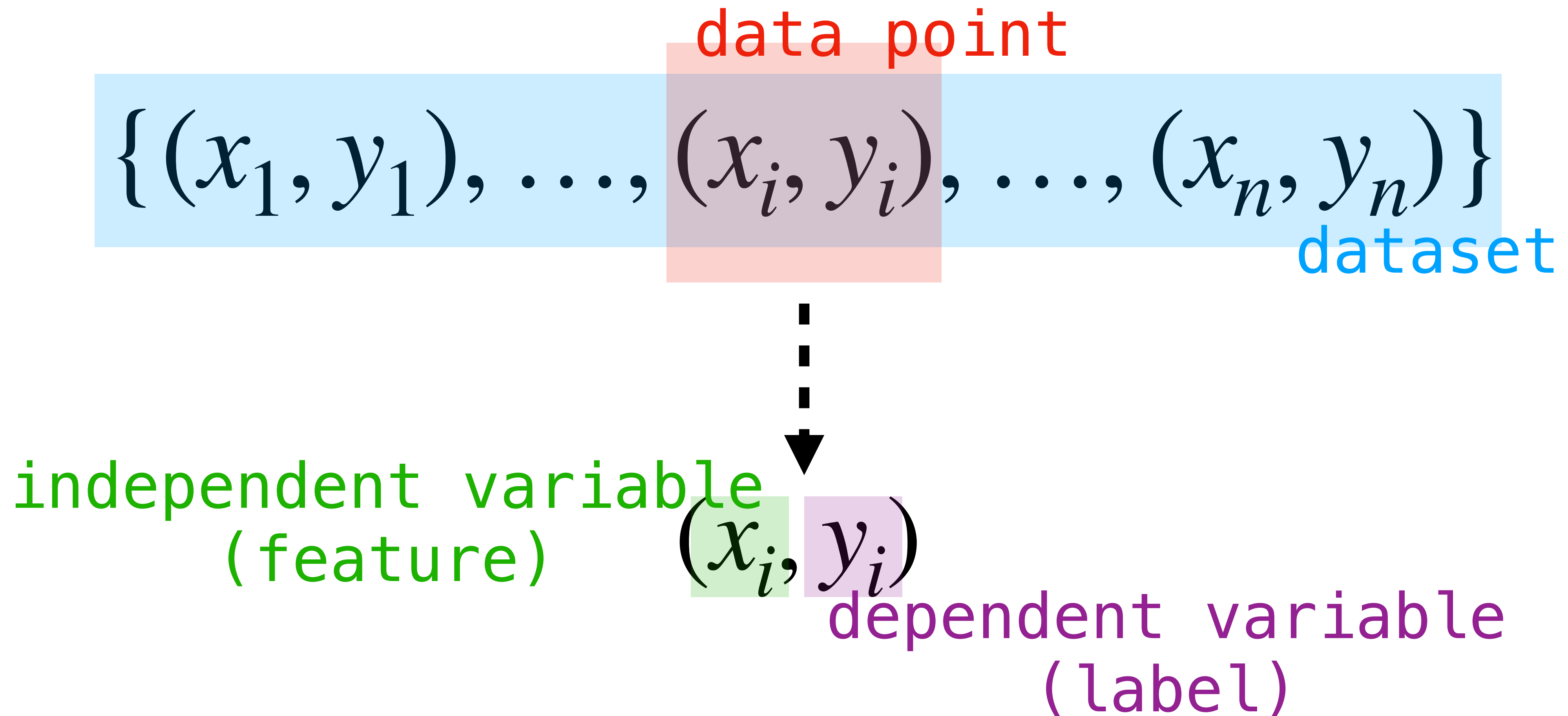
Terminology: Datasets



Terminology: Datasets



Terminology: Datasets



Terminology: Models

$$f(x) = \beta_0 + \beta_1 x$$

Terminology: Models

$$f(x) = \beta_0 + \beta_1 x$$

model

Terminology: Models

model parameters/
regression coefficients

$$f(x) = \beta_0 + \beta_1 x$$

model

Terminology: Least-Squares Error

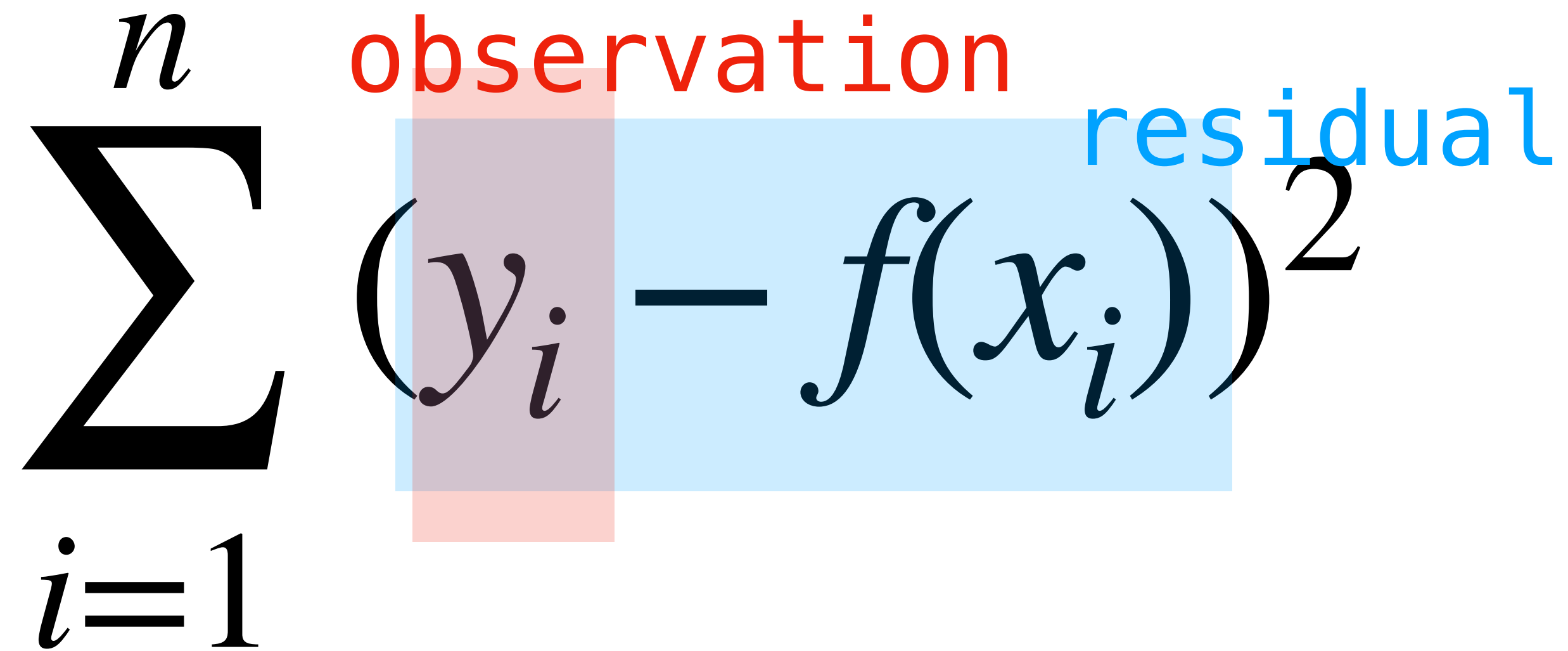
$$\sum_{i=1}^n (y_i - f(x_i))^2$$

Terminology: Least-Squares Error

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

residual

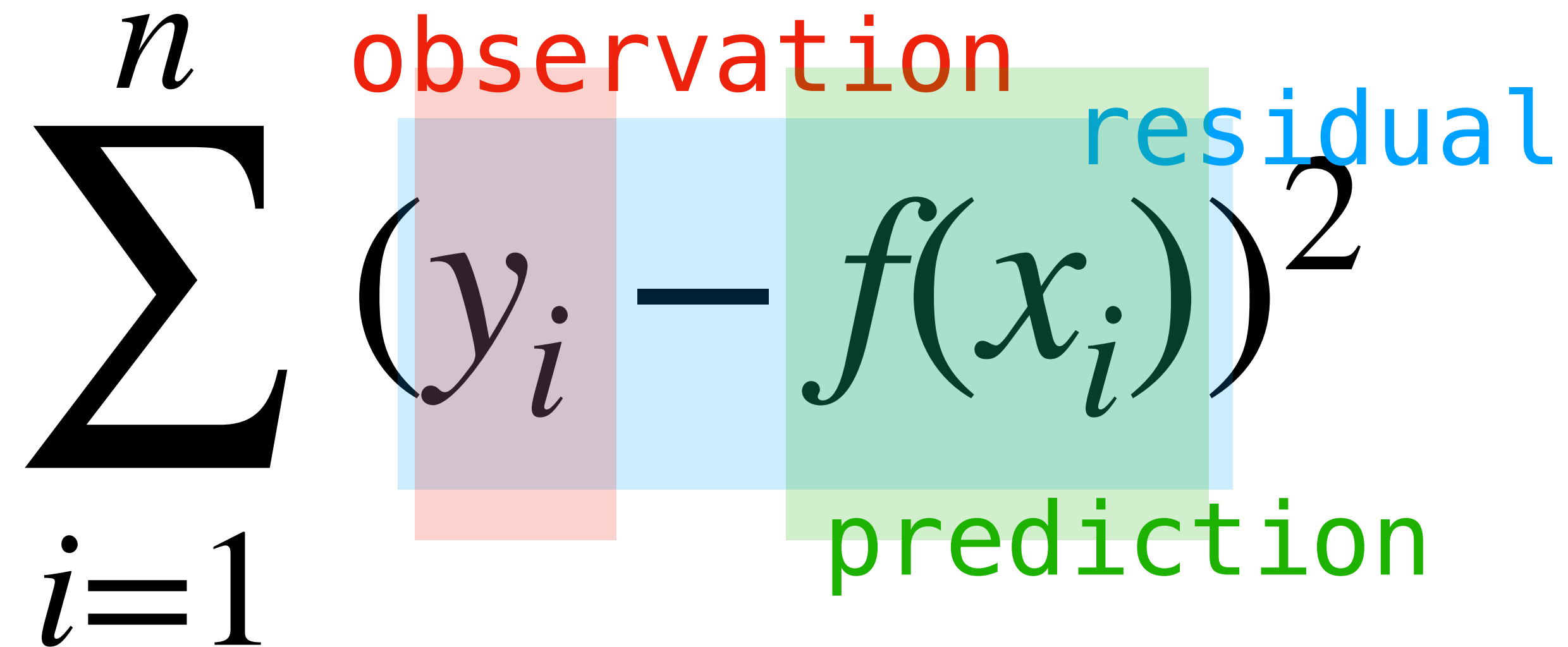
Terminology: Least-Squares Error



The diagram illustrates the least-squares error formula with two overlapping highlighted regions. A light red rectangular region highlights the term y_i , which is labeled "observation" in red text above it. A light blue rectangular region highlights the term $f(x_i)$, which is labeled "residual" in blue text above it. The entire expression is enclosed in large parentheses with a superscript of 2. The summation symbol $\sum$ is positioned to the left of the parentheses, with the index $i=1$ below it and the total number of observations n above it.

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

Terminology: Least-Squares Error



The diagram illustrates the least-squares error formula with color-coded components. The summation symbol $\sum$ is on the left, with the index $i=1$ below it and the total count n above it. The formula is $(y_i - f(x_i))^2$. The term y_i is enclosed in a light red box labeled "observation" in red text above it. The term $f(x_i)$ is enclosed in a light green box labeled "prediction" in green text below it. The entire expression $(y_i - f(x_i))^2$ is enclosed in a light blue box labeled "residual" in blue text above it.

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

How to: Finding the Least Squares Line

$$\beta_1 = \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2} \quad \beta_0 = \frac{\sum_{i=1}^n y_i - \beta_1 \sum_{i=1}^n x_i}{n}$$

How to: Finding the Least Squares Line

$$\beta_1 = \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2} \quad \beta_0 = \frac{\sum_{i=1}^n y_i - \beta_1 \sum_{i=1}^n x_i}{n}$$

Q. Find the least squares line for the dataset $\{(x_1, y_1), \dots, (x_n, y_n)\}$

How to: Finding the Least Squares Line

$$\beta_1 = \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2} \quad \beta_0 = \frac{\sum_{i=1}^n y_i - \beta_1 \sum_{i=1}^n x_i}{n}$$

Q. Find the least squares line for the dataset $\{(x_1, y_1), \dots, (x_n, y_n)\}$

A (First attempt). Use these equations...

How to: Finding the Least Squares Line

Don't memorize these

$$\beta_1 = \frac{\sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right) / n}{\sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2 / n} \quad \beta_0 = \frac{\sum_{i=1}^n y_i - \beta_1 \sum_{i=1}^n x_i}{n}$$

Q. Find the least squares line for the dataset $\{(x_1, y_1), \dots, (x_n, y_n)\}$

A (First attempt). Use these equations...

An Observation

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^n ((A\mathbf{x})_i - \mathbf{b}_i)^2$$

An Observation

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

minimize for least-squares line

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^n ((A\mathbf{x})_i - \mathbf{b}_i)^2$$

minimize for least-squares method

An Observation

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

minimize for least-squares line

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^n ((A\mathbf{x})_i - \mathbf{b}_i)^2$$

minimize for least-squares method

These expressions look very similar

An Observation

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

minimize for least-squares line

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \sum_{i=1}^n ((A\mathbf{x})_i - \mathbf{b}_i)^2$$

minimize for least-squares method

These expressions look very similar

Can we *design* a matrix where finding a least squares solution gives us a least squares line?

A Least Squares Problem

$$\beta_0 + \beta_1 x_1 = y_1$$

$$\beta_0 + \beta_1 x_2 = y_2$$

$$\vdots$$

$$\beta_0 + \beta_1 x_n = y_n$$

A Least Squares Problem

In the "ideal" world, we could find β_0 and β_1 such that all of these equations hold

$$\beta_0 + \beta_1 x_1 = y_1$$

$$\beta_0 + \beta_1 x_2 = y_2$$

$$\vdots$$

$$\beta_0 + \beta_1 x_n = y_n$$

A Least Squares Problem

In the "ideal" world, we could find β_0 and β_1 such that all of these equations hold

$$\beta_0 + \beta_1 x_1 = y_1$$

$$\beta_0 + \beta_1 x_2 = y_2$$

$$\vdots$$

This would mean **all the points already lie on a single line**

$$\beta_0 + \beta_1 x_n = y_n$$

A Least Squares Problem

In the "ideal" world, we could find β_0 and β_1 such that all of these equations hold

This would mean **all the points already lie on a single line**

$$1\beta_0 + x_1\beta_1 = y_1$$

$$\beta_0 + \beta_1 x_1 = y_1$$

$$\beta_0 + \beta_1 x_2 = y_2$$

$$\vdots$$

$$\beta_0 + \beta_1 x_n = y_n$$

This is a linear system in the variables β_0 and β_1

A Least Squares Problem

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

A Least Squares Problem

In the "ideal" world,
this matrix equation
has a solution

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

A Least Squares Problem

In the "ideal" world,
*this matrix equation
has a solution*

In reality it won't
have a solution, **but we
can find an approximate
solution**

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

A Least Squares Problem

$$\begin{bmatrix} 1 & \overset{\textcolor{red}{X}}{x_1} \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \overset{\textcolor{red}{\vec{\beta}}}{\beta_0} \\ \beta_1 \end{bmatrix} = \begin{bmatrix} \overset{\textcolor{red}{y}}{y_1} \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$\|X\vec{\beta} - \overset{\textcolor{red}{obs. \text{ val.}}}{y}\|^2 = \sum_{i=1}^n ((\beta_0 + \beta_1 x_i) - y_i)^2$$

A Least Squares Problem

$$\begin{bmatrix} 1 & \overset{\textcolor{red}{X}}{x_1} \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \overset{\textcolor{red}{\vec{\beta}}}{\beta_0} \\ \beta_1 \end{bmatrix} = \begin{bmatrix} \overset{\textcolor{red}{y}}{y_1} \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$\|X\vec{\beta} - \mathbf{y}\|^2 = \sum_{i=1}^n ((\beta_0 + \beta_1 x_i) - y_i)^2$$

The sum of squares of residuals is the squared distances between $X\beta$ and $\mathbf{y}$

A Least Squares Problem

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$\|X\vec{\beta} - \mathbf{y}\|^2 = \sum_{i=1}^n ((\beta_0 + \beta_1 x_i) - y_i)^2$$

The sum of squares of residuals is the squared distances between $X\beta$ and $\mathbf{y}$

Least squares solutions to this system give us parameters for least squares lines

Recall: The Normal Equations

Recall: The Normal Equations

Thm. The set of least-squares solutions of $A\mathbf{x} = \mathbf{b}$ is the same as the set of solutions to

$$A^T A \mathbf{x} = A^T \mathbf{b}$$

Recall: The Normal Equations

Thm. The set of least-squares solutions of $A\mathbf{x} = \mathbf{b}$ is the same as the set of solutions to

$$A^T A\mathbf{x} = A^T \mathbf{b}$$

In particular, this set of solutions is nonempty

Recall: Unique Least Squares Solutions

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$$

If A has LI columns, then its unique least squares solution is defined as above

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

How To: Least Squares Line

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

How To: Least Squares Line

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Q. Find the least squares line for the dataset $\{(x_1, y_1), \dots, (x_n, y_n)\}$

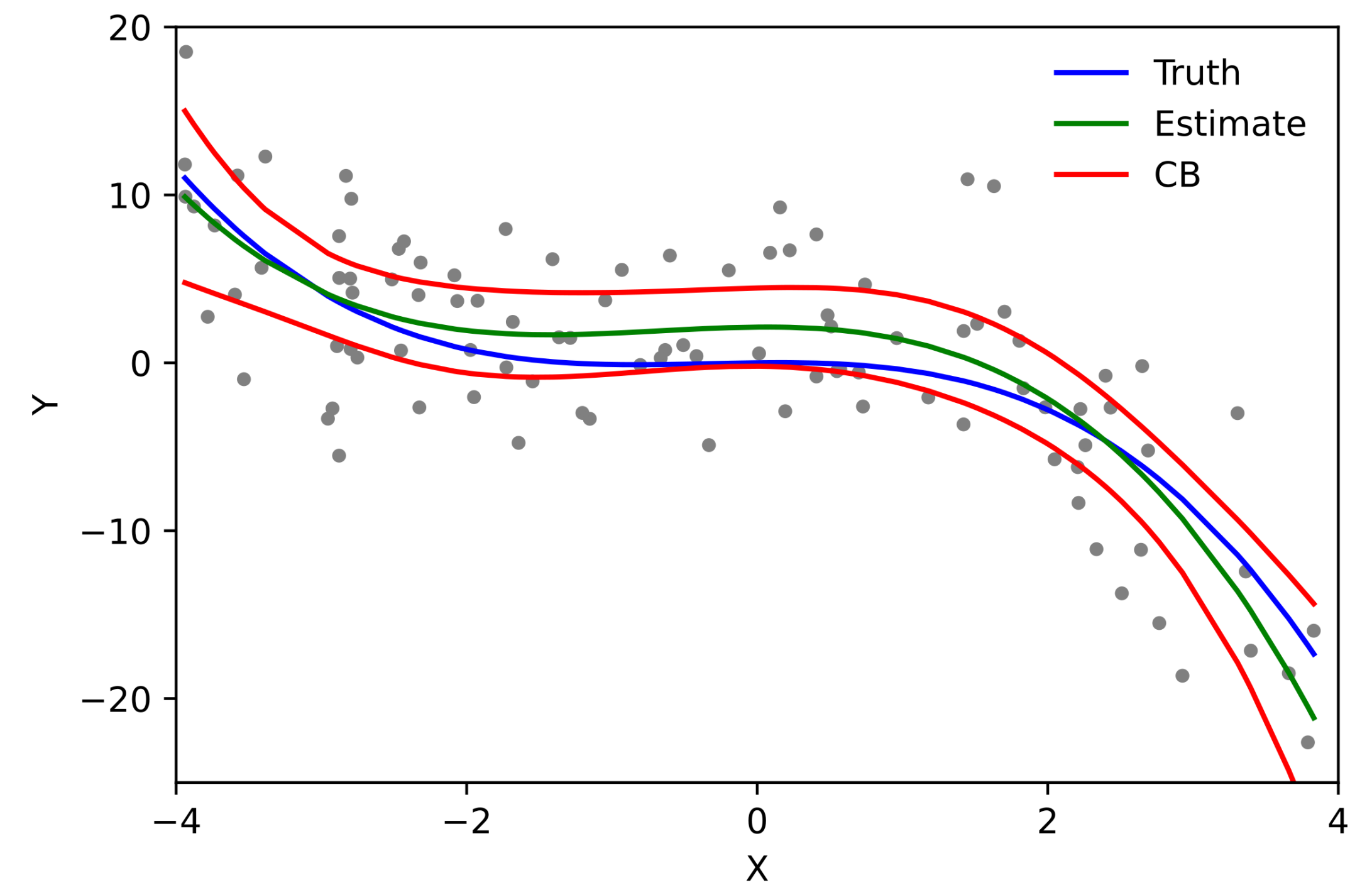
How To: Least Squares Line

$$\begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Q. Find the least squares line for the dataset $\{(x_1, y_1), \dots, (x_n, y_n)\}$

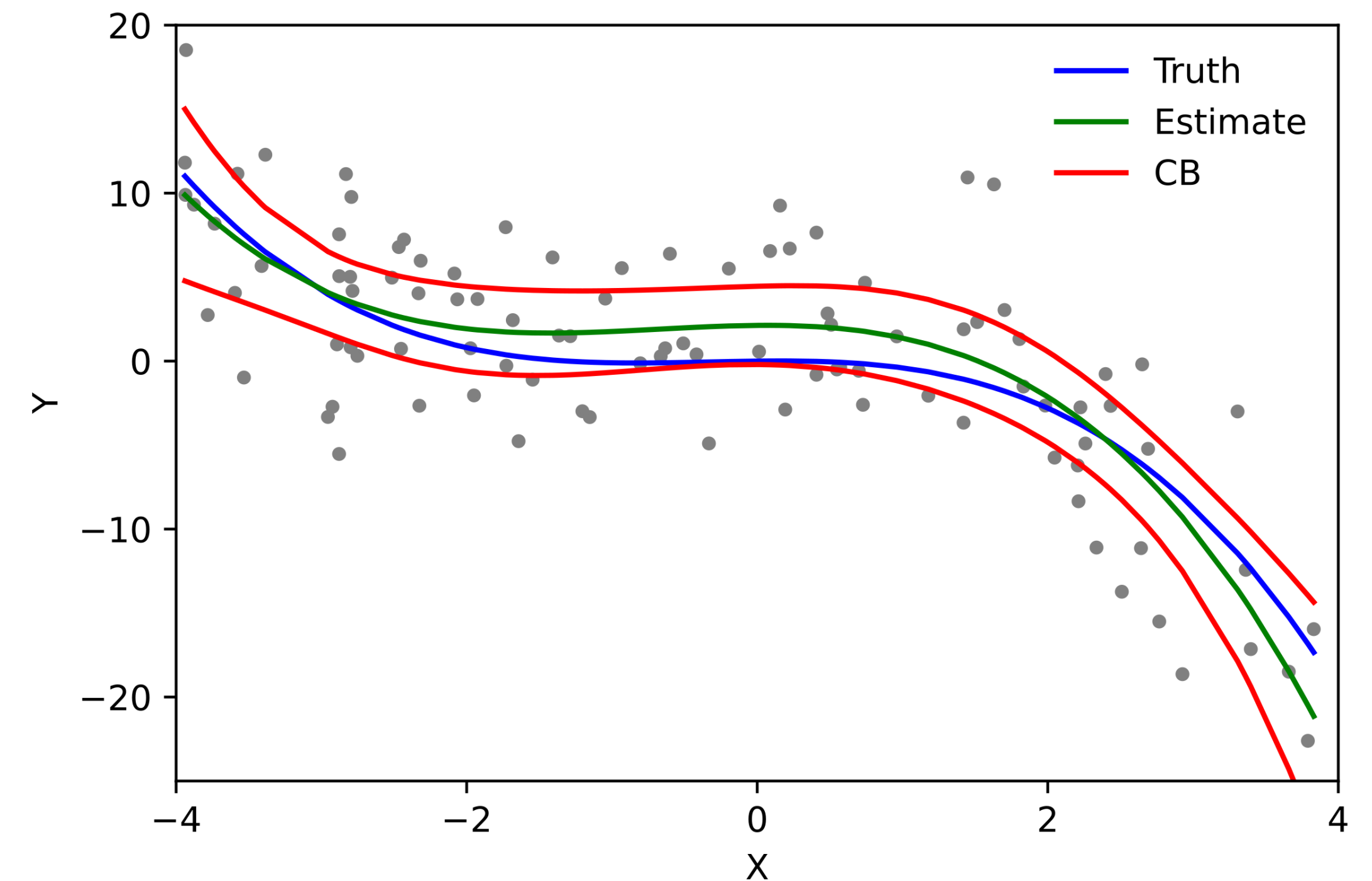
A. Find the least squares solution to the above equation

General Regression



General Regression

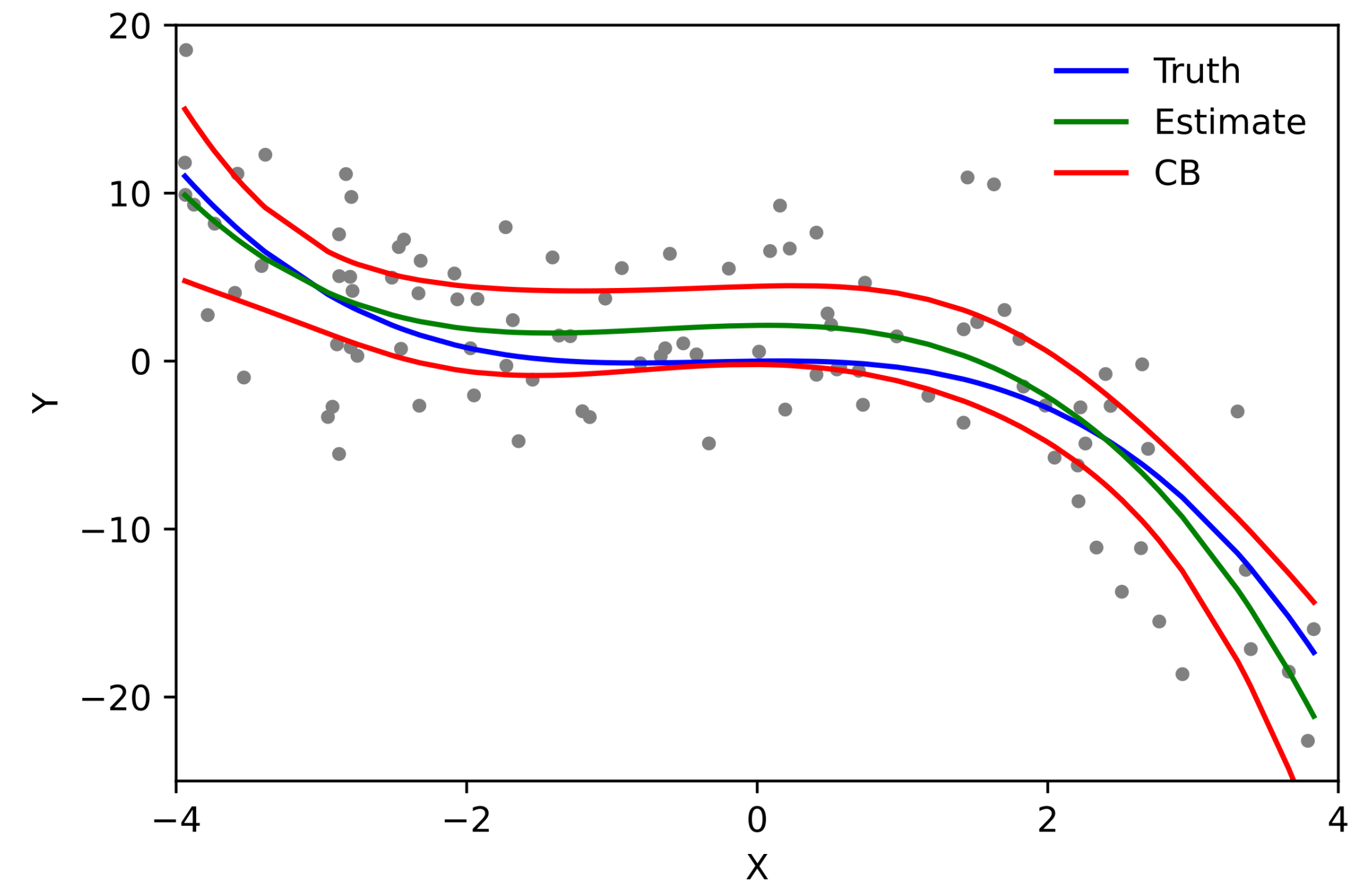
Regression is estimating the relationship between ind. and dep. variables in a dataset



General Regression

Regression is estimating the relationship between ind. and dep. variables in a dataset

What we are estimating is a *mathematical function*

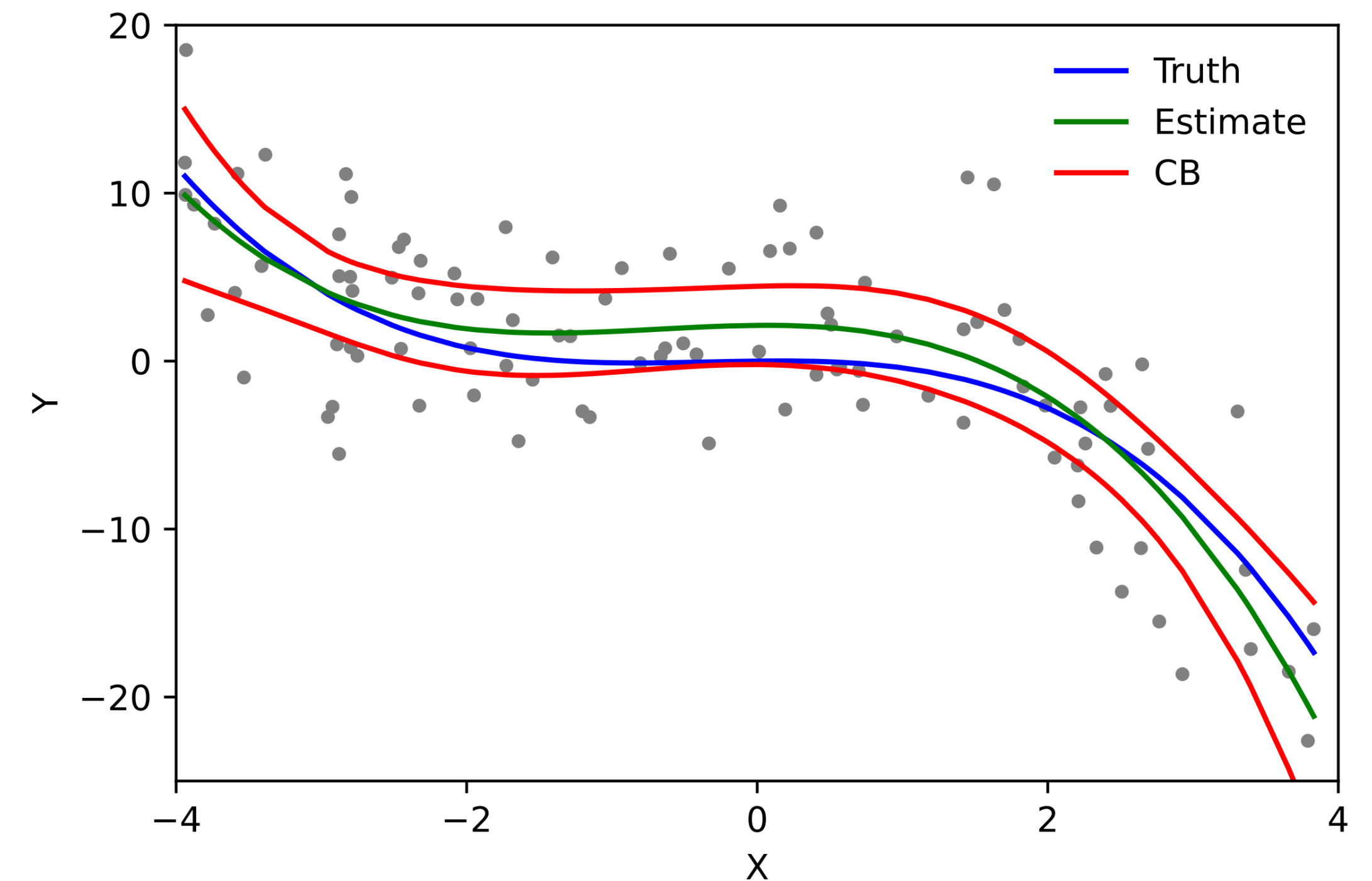


General Regression

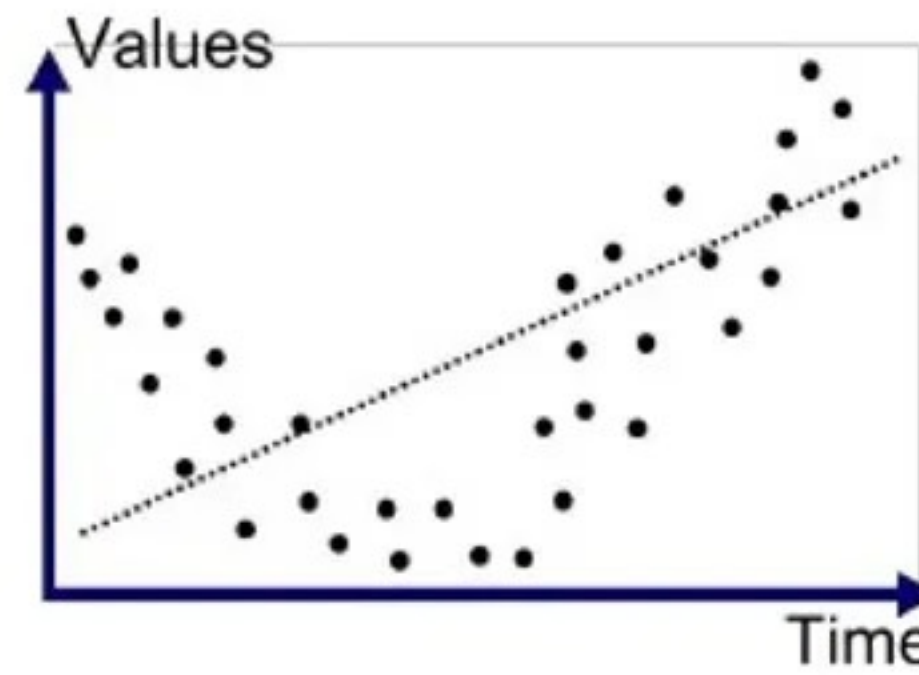
Regression is estimating the relationship between ind. and dep. variables in a dataset

What we are estimating is a *mathematical function*

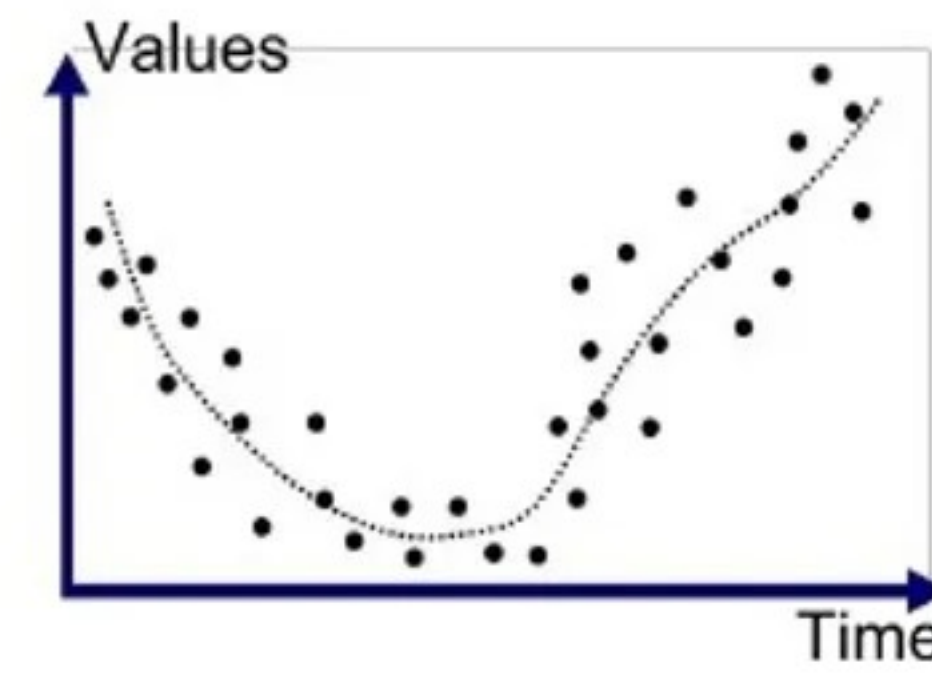
We think of the environment has providing us a function from our ind. variables to our dep. variables



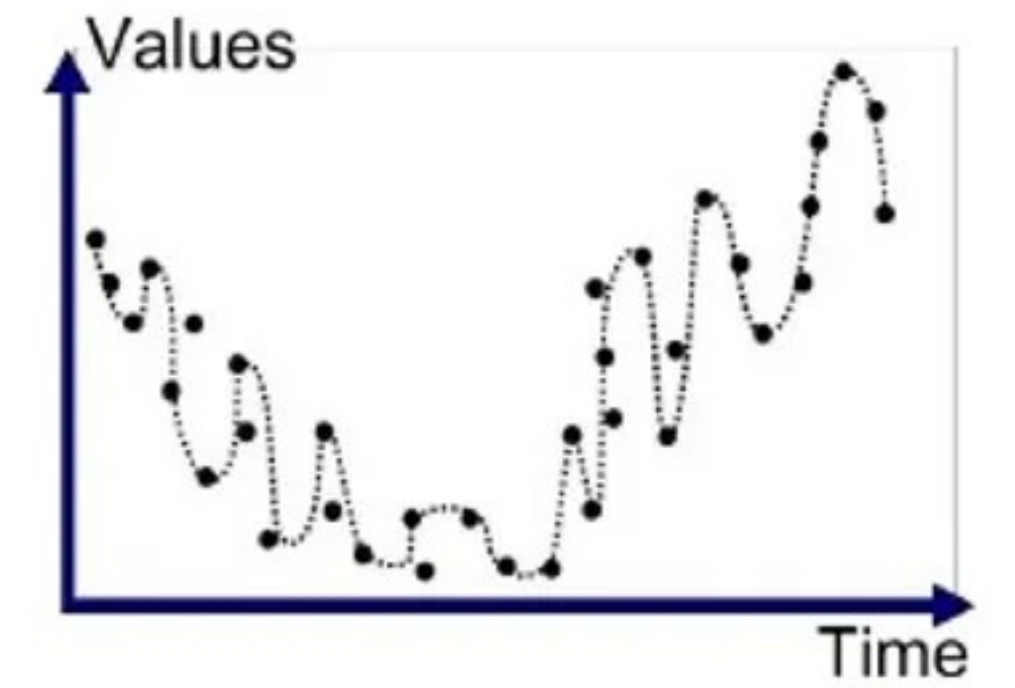
Models



Underfitted

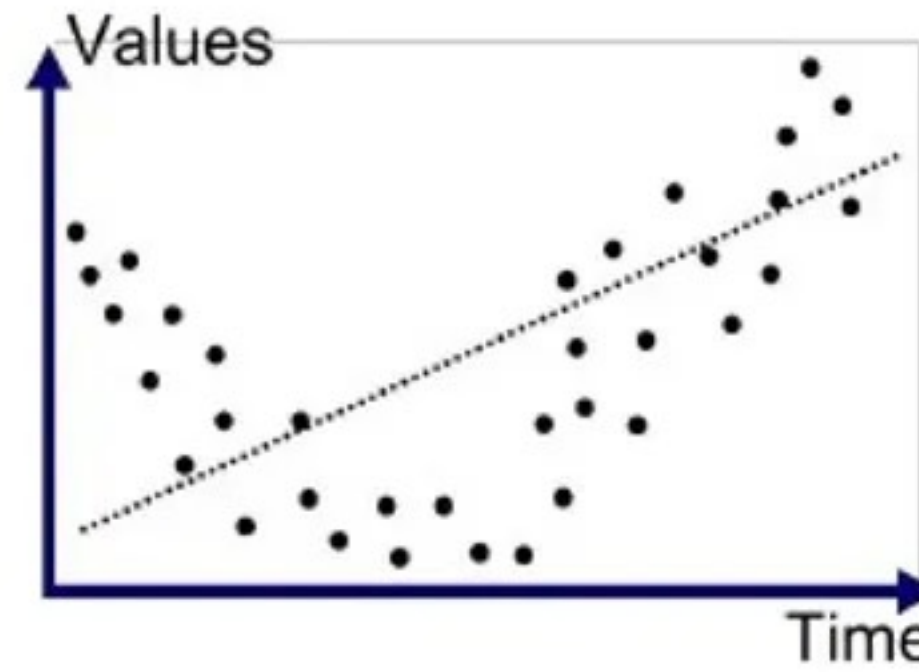


Good Fit/Robust

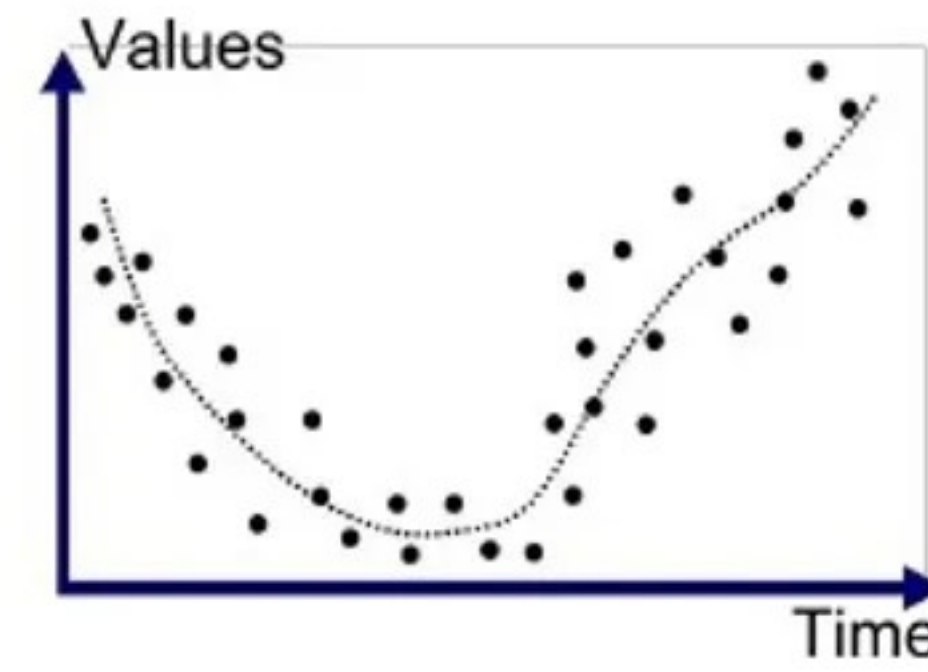


Overfitted

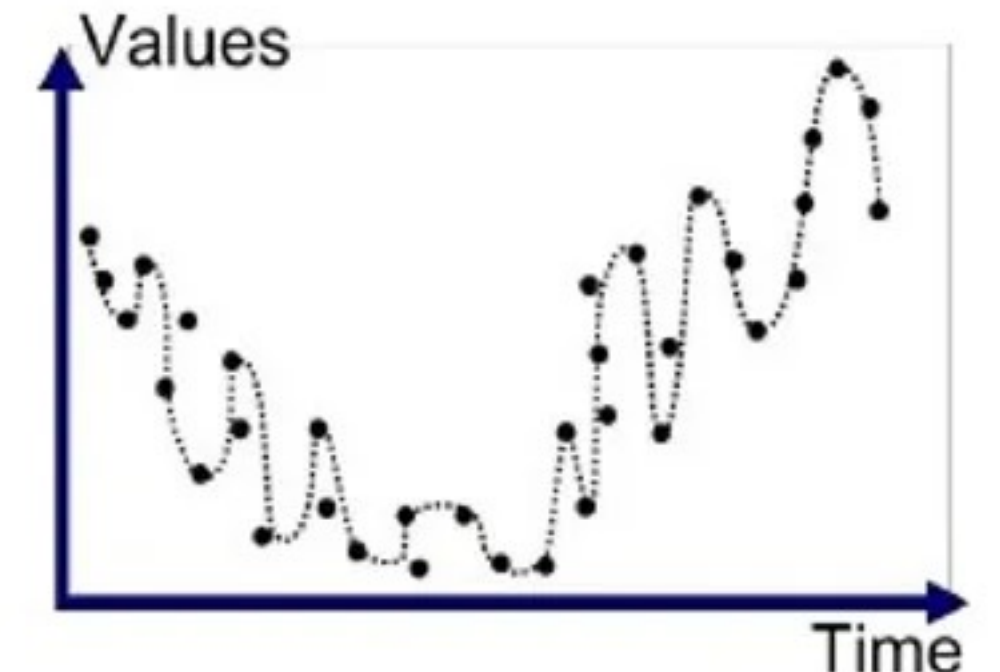
Models



Underfitted



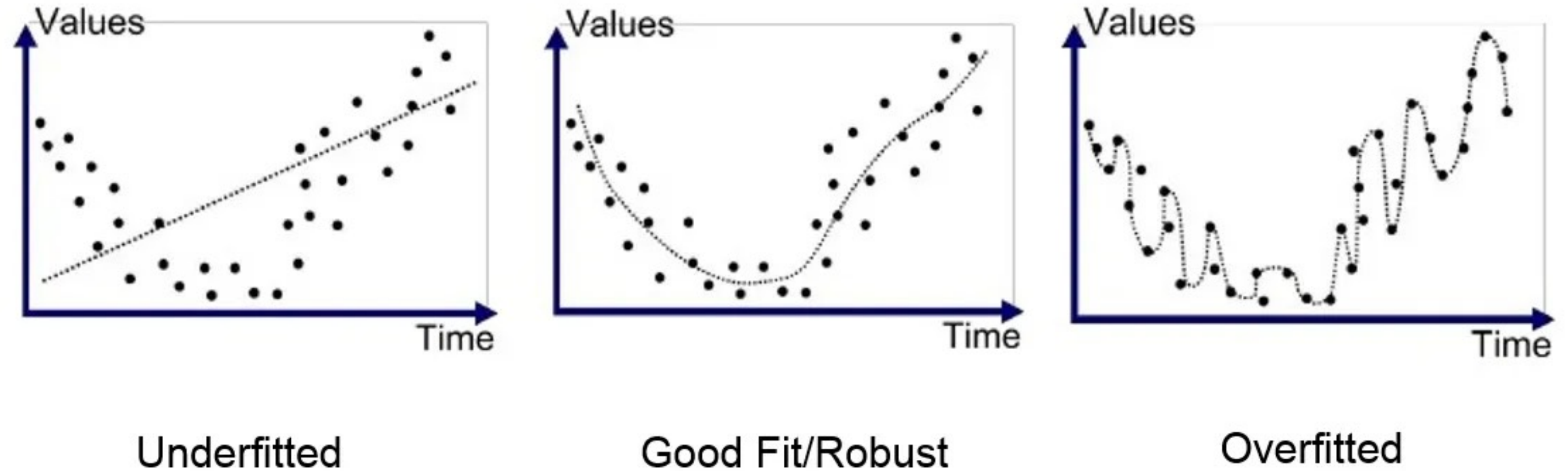
Good Fit/Robust



Overfitted

Therefore, a *model* is a mathematical function

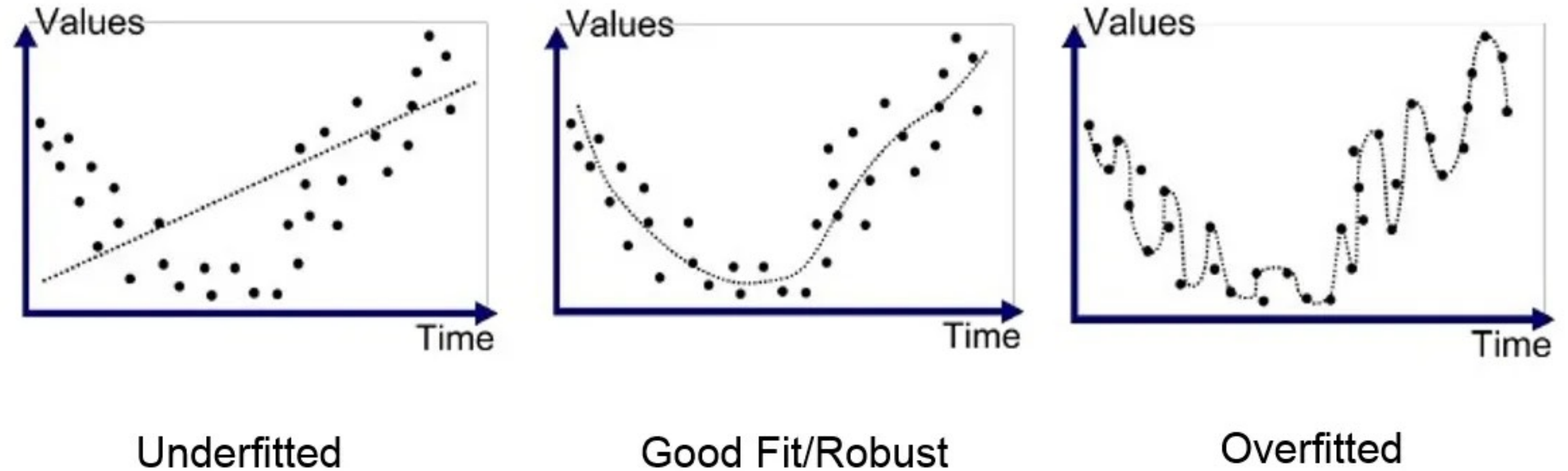
Models



Therefore, a *model* is a mathematical function

We're interested in mathematical functions that "correctly" model the data we've seen

Models

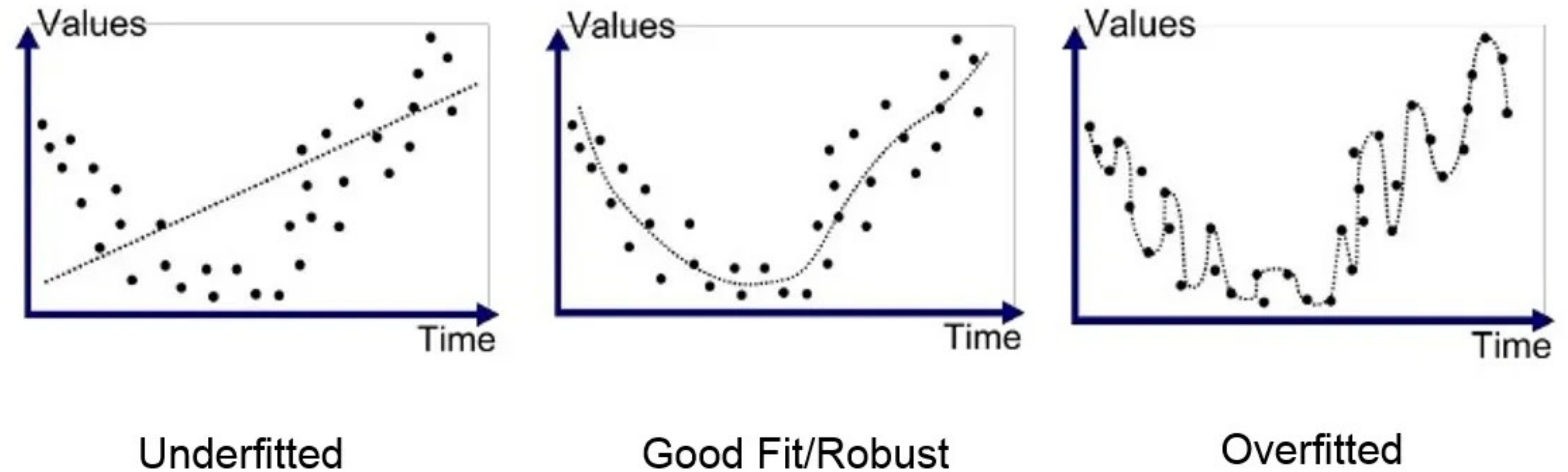


Therefore, a *model* is a mathematical function

We're interested in mathematical functions that "correctly" model the data we've seen

But this would be boring if we *just* modeled data we've seen

Models



Therefore, a *model* is a mathematical function

We're interested in mathematical functions that "correctly" model the data we've seen

But this would be boring if we *just* modeled data we've seen

(Advanced) We pick models from weaker classes of functions so that they are more robust when we **predict** values using the model

How To: Prediction

How To: Prediction

Q. Given $\{(x_1, y_1), \dots, (x_k, y_k)\}$ use the line of best fit to predict the value of y' for the input x'

How To: Prediction

Q. Given $\{(x_1, y_1), \dots, (x_k, y_k)\}$ use the line of best fit to predict the value of y' for the input x'

A. Find the best fit line $f(x) = \beta_0 + \beta_1 x$. The predicted value of x' is $f(x')$

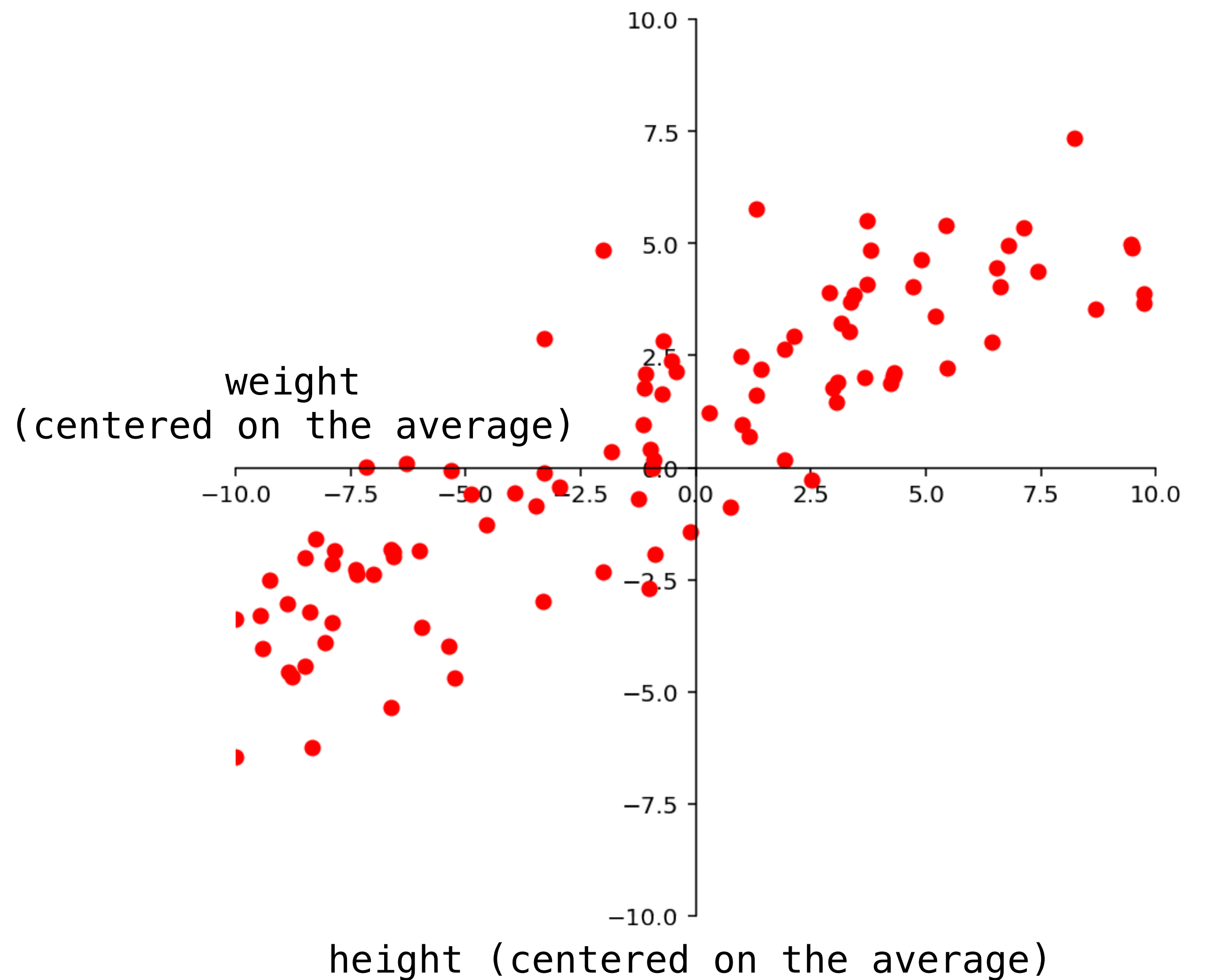
How To: Prediction

Q. Given $\{(x_1, y_1), \dots, (x_k, y_k)\}$ use the line of best fit to predict the value of y' for the input x'

A. Find the best fit line $f(x) = \beta_0 + \beta_1 x$. The predicted value of x' is $f(x')$

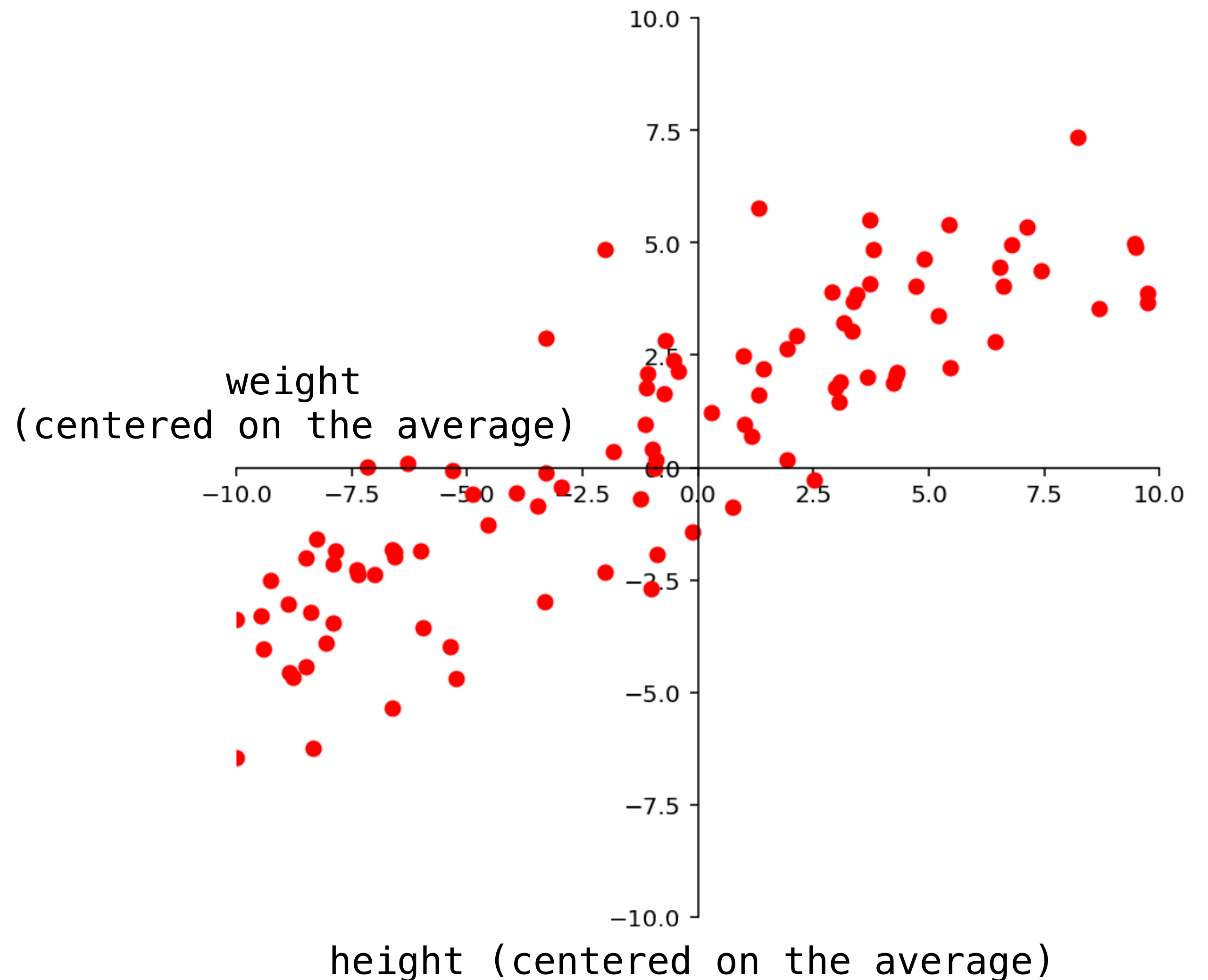
This generalizes to any model fitting problem

Example: Height from Weight



Example: Height from Weight

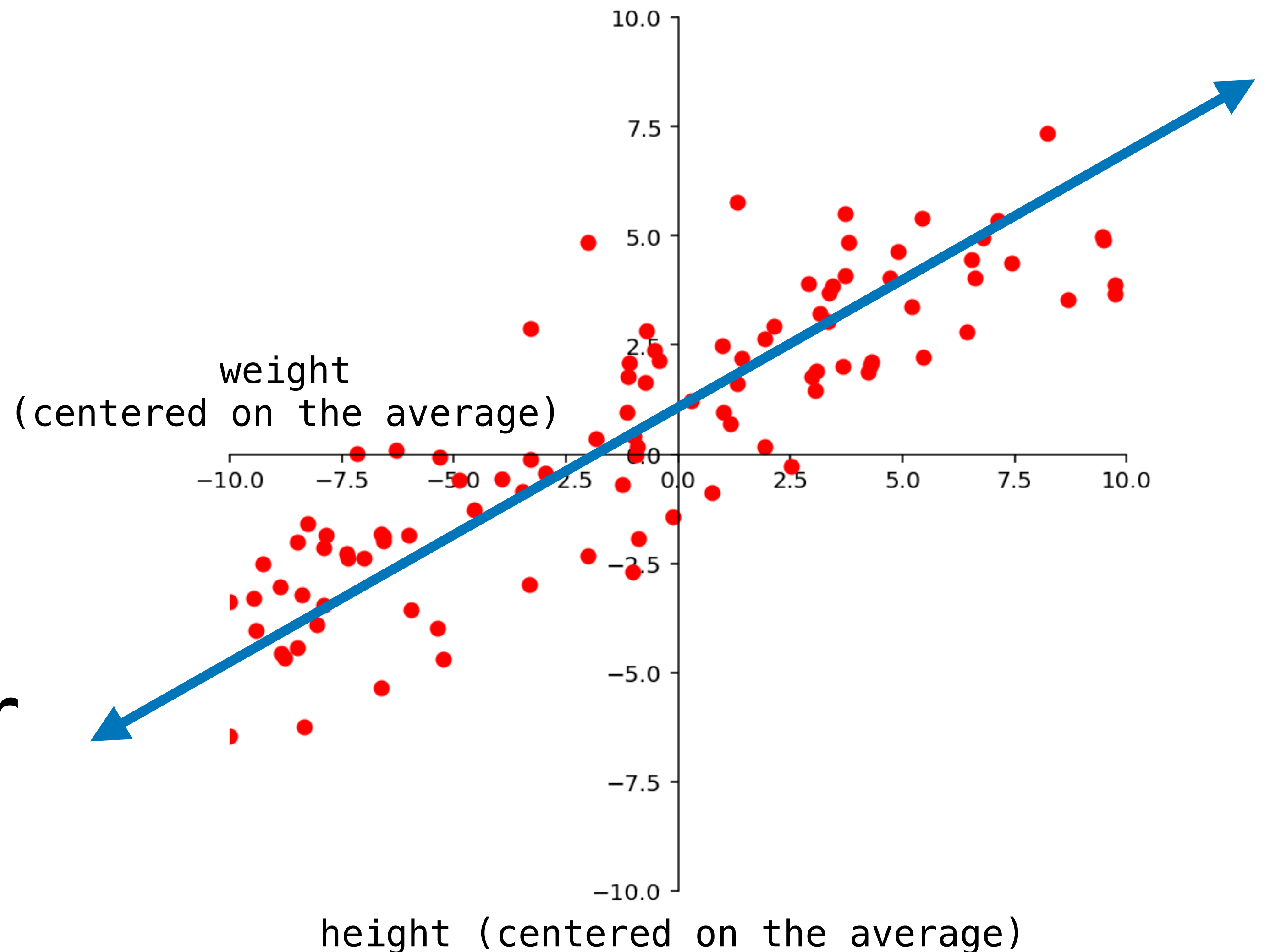
Suppose we know that person X weighs 150lb.
How would we guess the height of person X ?



Example: Height from Weight

Suppose we know that person X weighs 150lb. *How would we guess the height of person X ?*

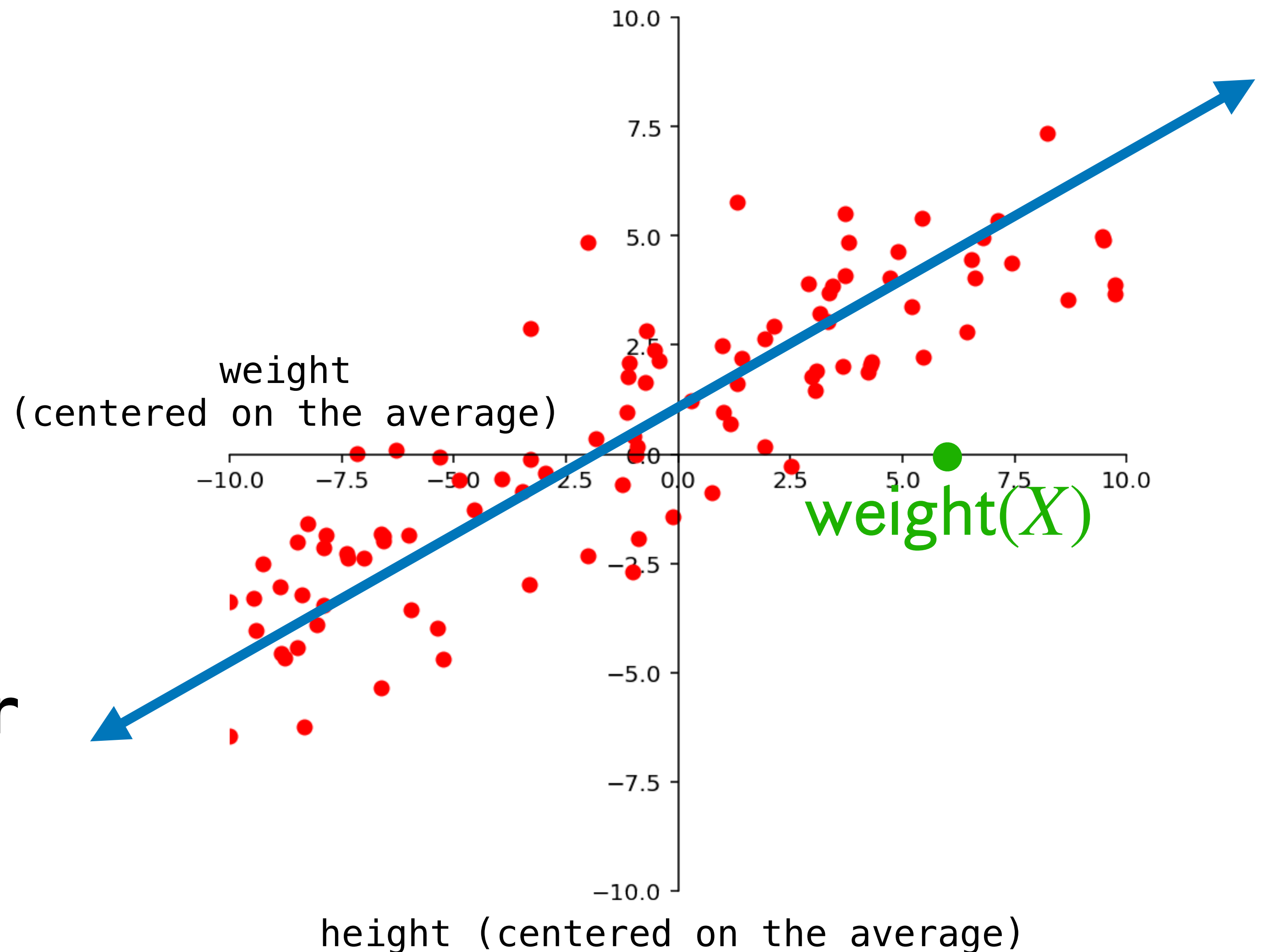
If we know population data, then we can **find the line of best fit for that data** and then use that function



Example: Height from Weight

Suppose we know that person X weighs 150lb. *How would we guess the height of person X ?*

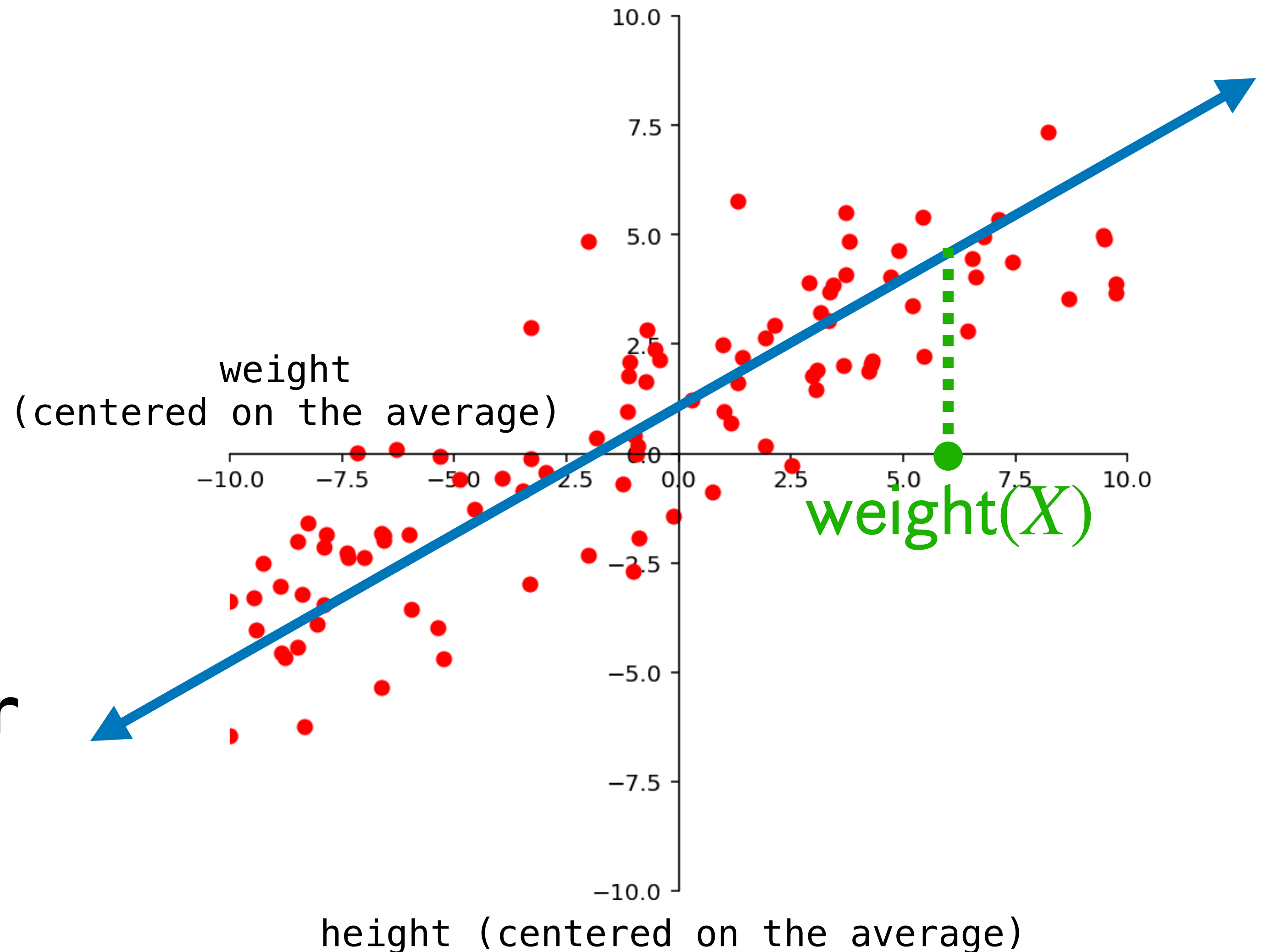
If we know population data, then we can **find the line of best fit for that data** and then use that function



Example: Height from Weight

Suppose we know that person X weighs 150lb. *How would we guess the height of person X ?*

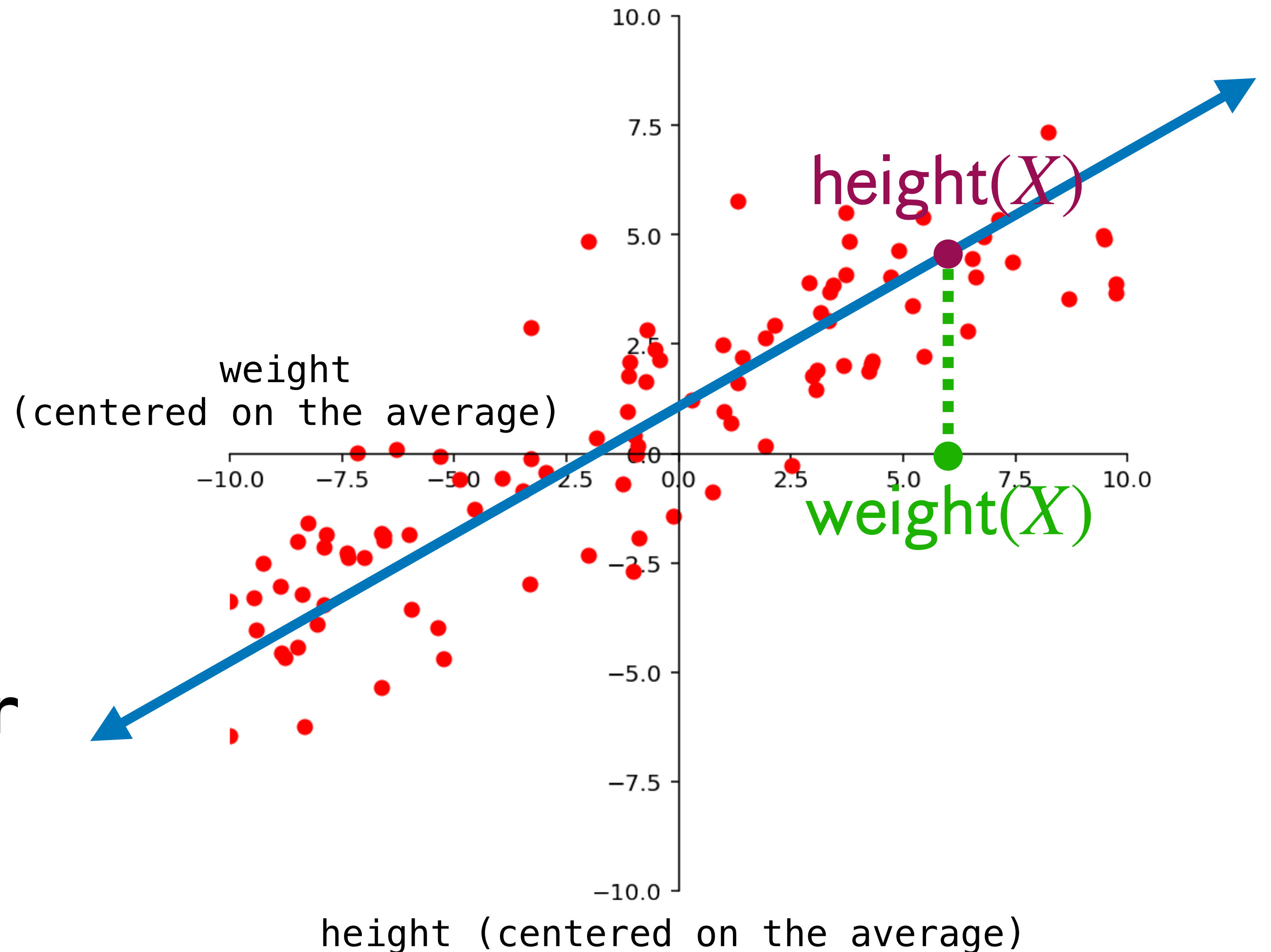
If we know population data, then we can **find the line of best fit for that data** and then use that function



Example: Height from Weight

Suppose we know that person X weighs 150lb. *How would we guess the height of person X ?*

If we know population data, then we can **find the line of best fit for that data** and then use that function



Question

Find the line of best fit for the dataset

$$\{(0,3), (1,1), (-1,1), (2,3)\}$$

by using the the least-squares method

If you have time, graph your result and use it to "predict" the corresponding value for the input 4

Answer

$$\{(0,3), (1,1), (-1,1), (2,3)\}$$

Linear Models and Least Squares Regression

"Vectors" of Generalization

"Vectors" of Generalization

1. What if we have *more than one* independent value?

"Vectors" of Generalization

1. What if we have *more than one* independent value?

multiple regression, (hyper)plane of best fit

"Vectors" of Generalization

1. What if we have *more than one* independent value?

multiple regression, (hyper)plane of best fit

1. What if our data is not *exactly* linear?

"Vectors" of Generalization

1. What if we have *more than one* independent value?

multiple regression, (hyper)plane of best fit

1. What if our data is not *exactly* linear?

e.g., polynomial regression

"Vectors" of Generalization

1. What if we have *more than one* independent value?

multiple regression, (hyper)plane of best fit

1. What if our data is not *exactly* linear?

e.g., polynomial regression

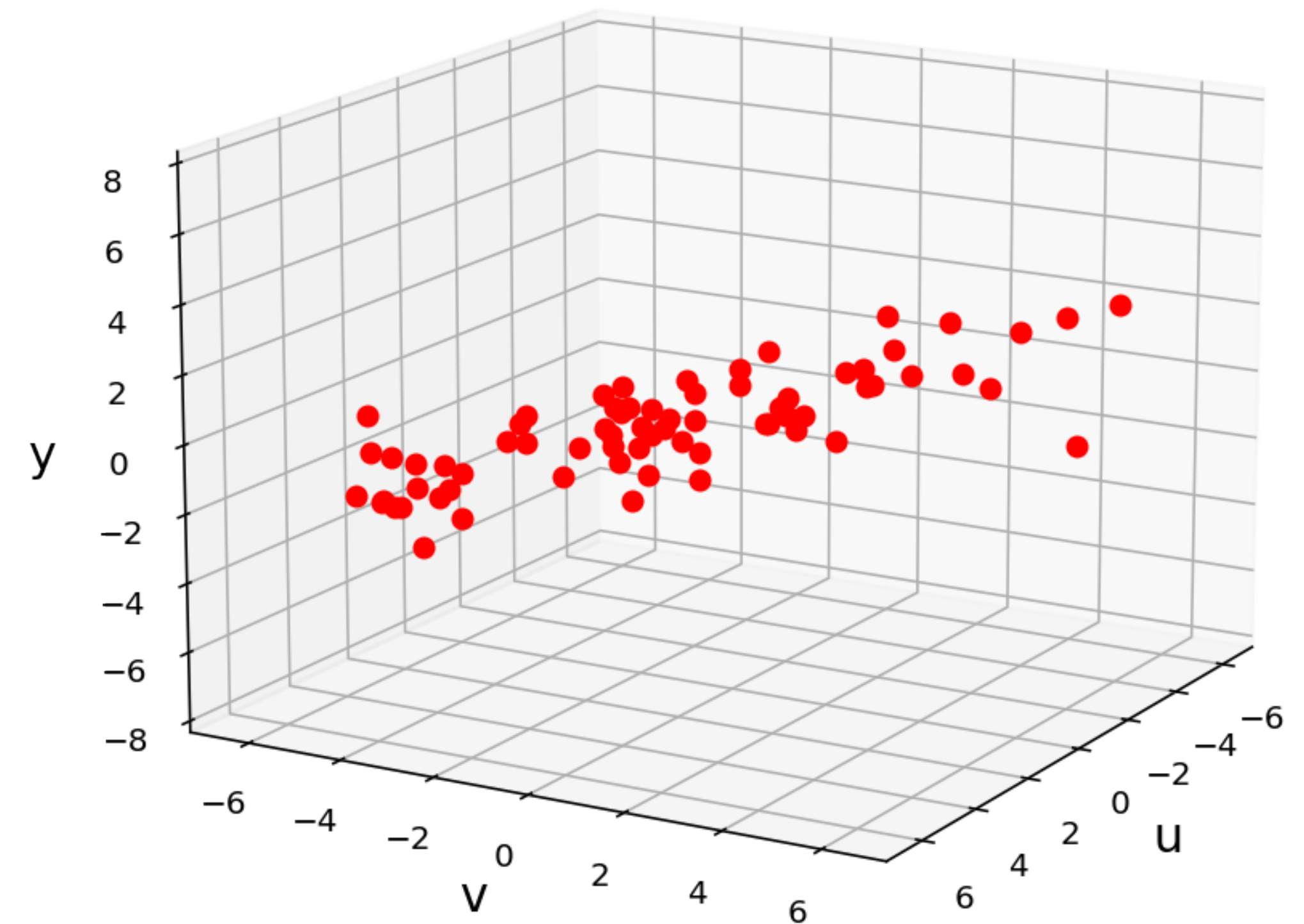
Example: Terrain Data

Figure 23.1

Terrain Data for Multiple Regression

Dataset: $\{(x_1, y_1, z_1), \dots, (x_k, y_k, z_k)\}$
where (x_i, y_i) is an longitude
and latitude and z_i is an
altitude

*Find the plane which
"best" fits the data*



Example: Terrain Data

Figure 23.2

Multiple Regression Fit to Data

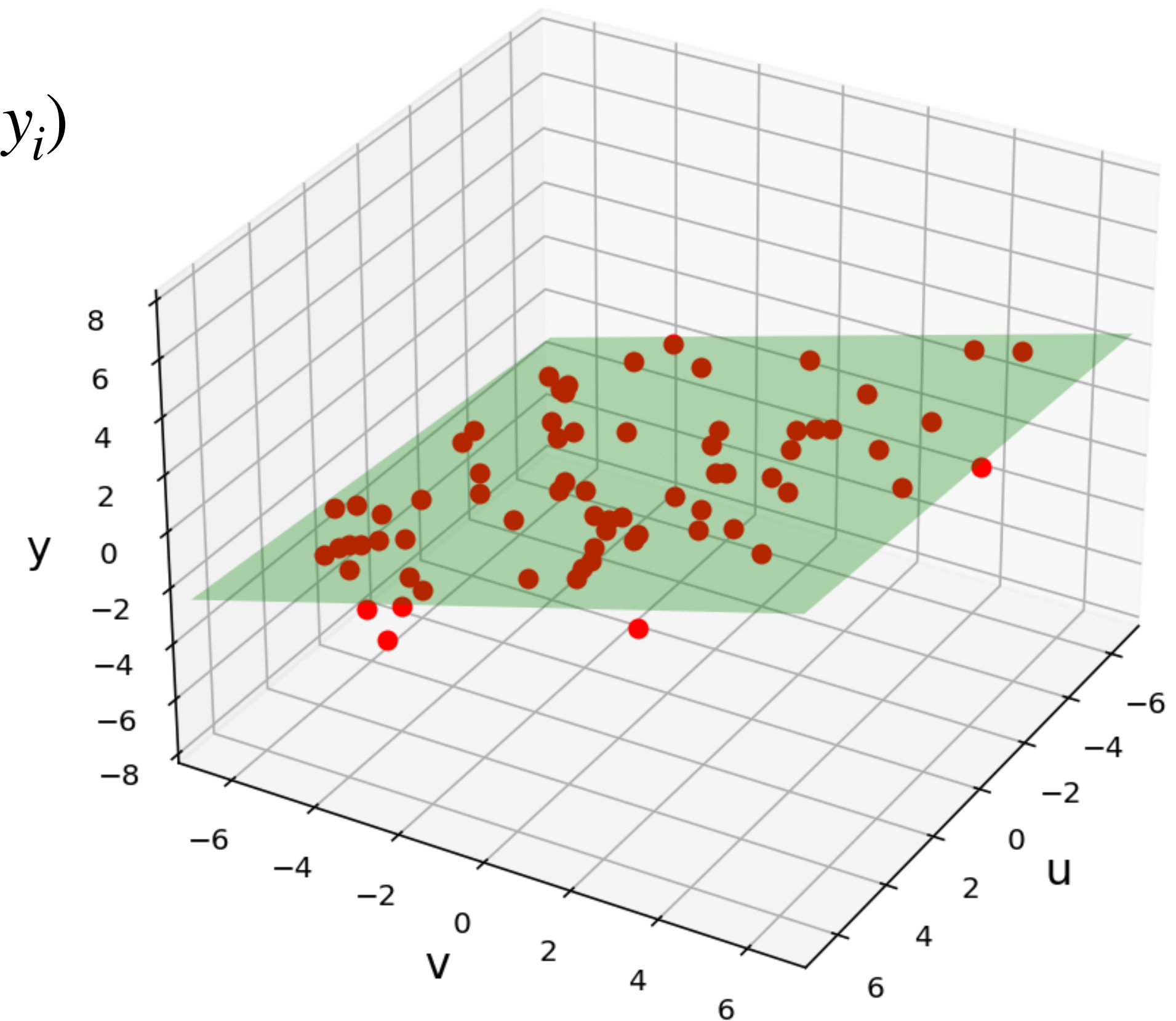
Dataset: $\{(x_1, y_1, z_1), \dots, (x_k, y_k, z_k)\}$ where (x_i, y_i) is an longitude and latitude and z_i is an altitude

Find $\beta_0, \beta_1, \beta_2$ such that

$$f(x, y) = \beta_0 + \beta_1 x + \beta_2 y$$

which minimizes

$$\sum_{i=1}^k (f(x_i, y_i) - z_i)^2$$



Example: Terrain Data

Figure 23.2

Multiple Regression Fit to Data

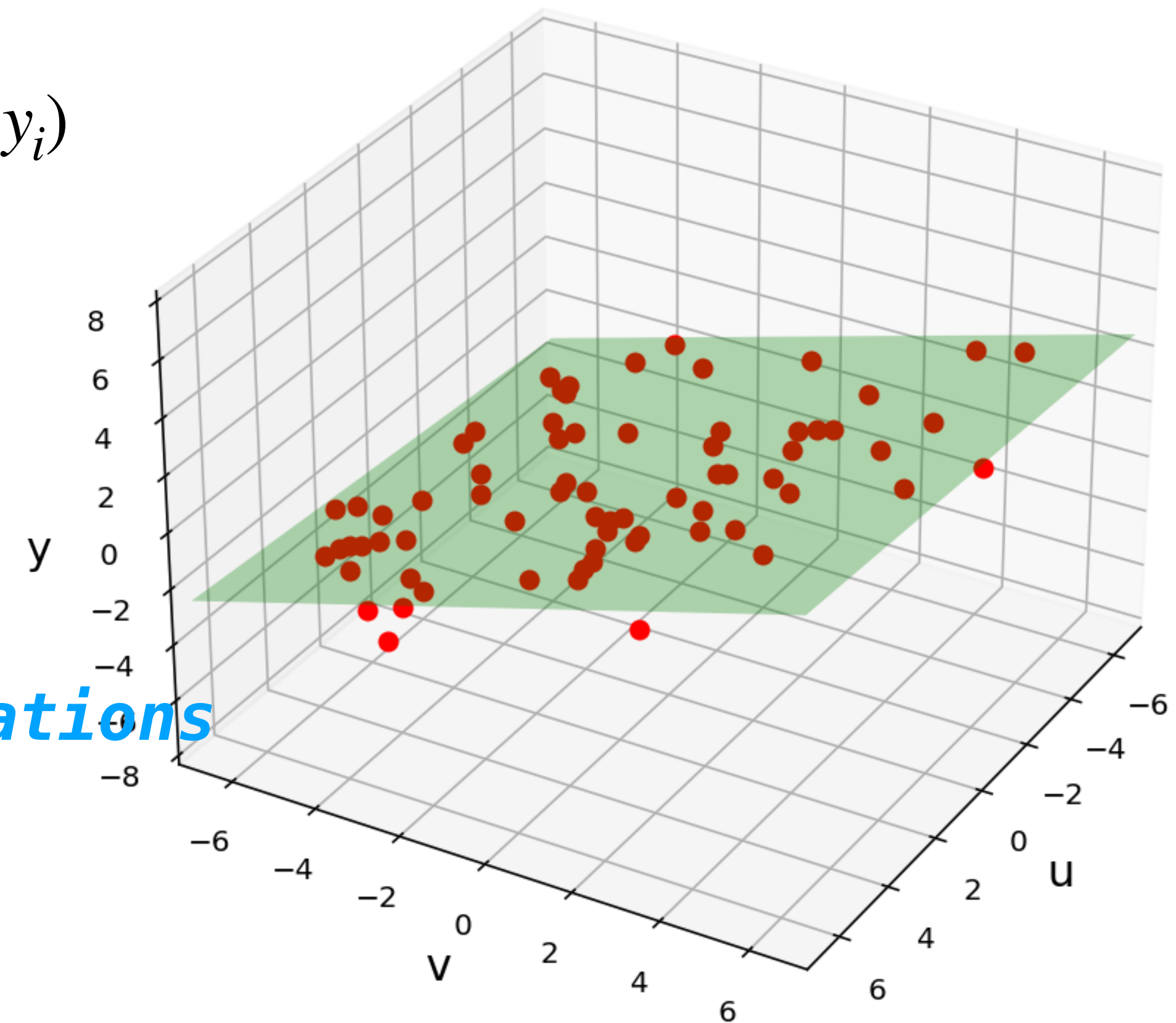
Dataset: $\{(x_1, y_1, z_1), \dots, (x_k, y_k, z_k)\}$ where (x_i, y_i) is an longitude and latitude and z_i is an altitude

Find $\beta_0, \beta_1, \beta_2$ such that

$$f(x, y) = \beta_0 + \beta_1 x + \beta_2 y$$

recall: planes are given by linear equations
which minimizes

$$\sum_{i=1}^k (f(x_i, y_i) - z_i)^2$$



Example: Terrain Data

Figure 23.2

Multiple Regression Fit to Data

Dataset: $\{(x_1, y_1, z_1), \dots, (x_k, y_k, z_k)\}$ where (x_i, y_i) is an longitude and latitude and z_i is an altitude

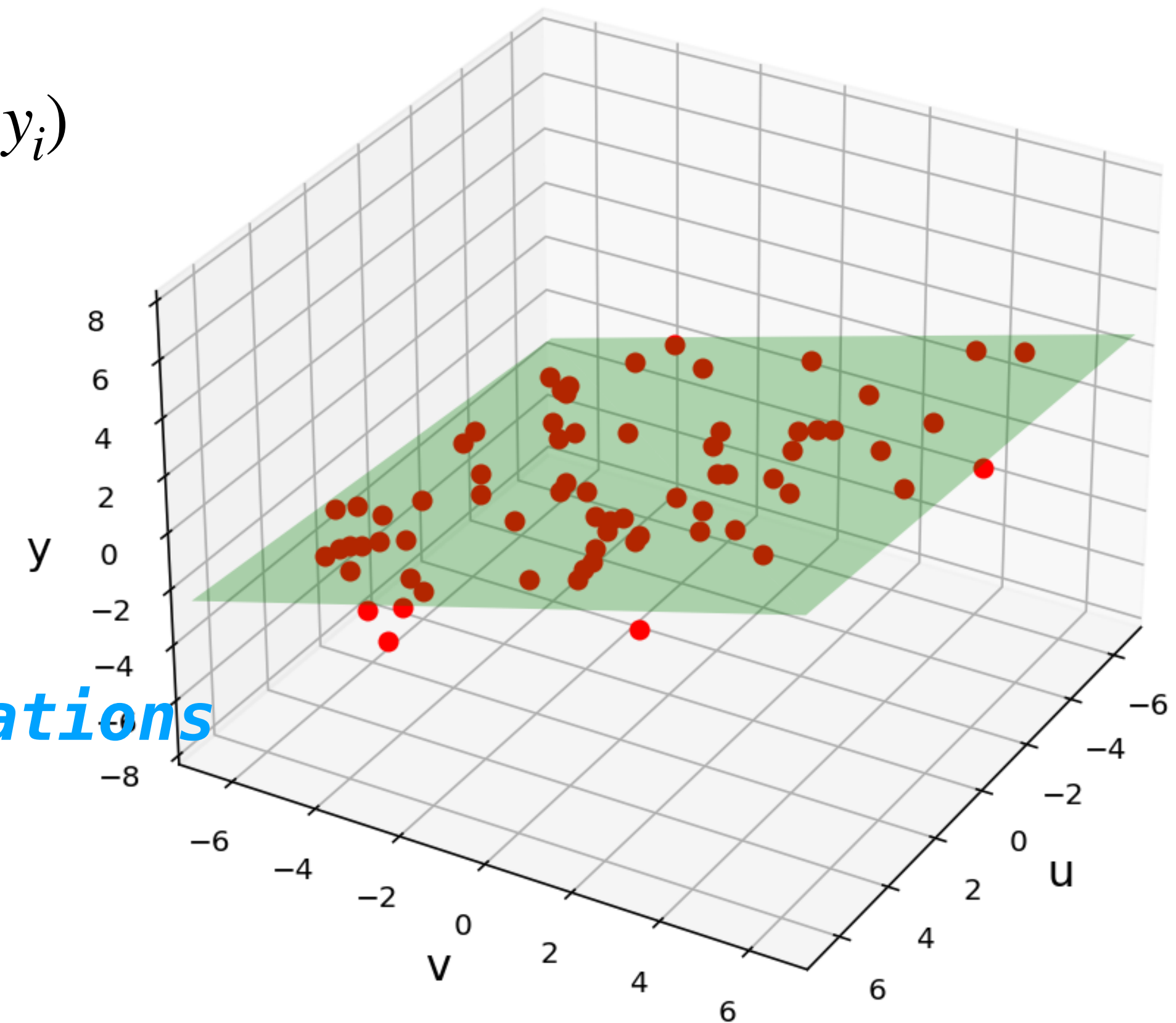
Find $\beta_0, \beta_1, \beta_2$ such that

$$f(x, y) = \beta_0 + \beta_1 x + \beta_2 y$$

recall: planes are given by linear equations
which minimizes

$$\sum_{i=1}^k (f(x_i, y_i) - z_i)^2$$

$f(x, y)$ is a good approximation of the altitude



Example: Terrain Data

Dataset: $\{(x_1, y_1, z_1), \dots, (x_k, y_k, z_k)\}$ where (x_i, y_i) is an longitude and latitude and z_i is an altitude

Find $\beta_0, \beta_1, \beta_2$ such that

$$f(x, y) = \beta_0 + \beta_1 x + \beta_2 y$$

which minimizes

$$\sum_{i=1}^k (f(x_i, y_i) - z_i)^2$$

$$\beta_0 + \beta_1 x_1 + \beta_2 y_1 = z_1$$

$$\beta_0 + \beta_1 x_2 + \beta_2 y_2 = z_2$$

$\vdots$

$$\beta_0 + \beta_1 x_k + \beta_2 y_k = z_k$$

Step 1: Set up an (almost assuredly inconsistent) system of linear equations in terms of the variables $\beta_0, \beta_1, \beta_2$

Example: Terrain Data

This is still linear in the β 's

Dataset: $\{(x_1, y_1, z_1), \dots, (x_k, y_k, z_k)\}$ where (x_i, y_i) is an longitude and latitude and z_i is an altitude

Find $\beta_0, \beta_1, \beta_2$ such that

$$f(x, y) = \beta_0 + \beta_1 x + \beta_2 y$$

which minimizes

$$\sum_{i=1}^k (f(x_i, y_i) - z_i)^2$$

$$\beta_0 + \beta_1 x_1 + \beta_2 y_1 = z_1$$

$$\beta_0 + \beta_1 x_2 + \beta_2 y_2 = z_2$$

$\vdots$

$$\beta_0 + \beta_1 x_k + \beta_2 y_k = z_k$$

Step 1: Set up an (almost assuredly inconsistent) system of linear equations in terms of the variables $\beta_0, \beta_1, \beta_2$

Example: Terrain Data

Dataset: $\{(x_1, y_1, z_1), \dots, (x_k, y_k, z_k)\}$ where (x_i, y_i) is an longitude and latitude and z_i is an altitude

Find $\beta_0, \beta_1, \beta_2$ such that

$$f(x, y) = \beta_0 + \beta_1 x + \beta_2 y$$

which minimizes

$$\sum_{i=1}^k (f(x_i, y_i) - z_i)^2$$

$$\begin{bmatrix} 1 & \overset{\textcolor{blue}{X}}{x_1} & y_1 \\ 1 & x_2 & y_2 \\ \vdots & \vdots & \vdots \\ 1 & x_k & y_k \end{bmatrix} \begin{bmatrix} \overset{\textcolor{blue}{\vec{\beta}}}{\beta_0} \\ \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} \overset{\textcolor{blue}{z}}{z_1} \\ z_2 \\ \vdots \\ z_k \end{bmatrix}$$

Step 2: Rewrite the system as a matrix equation

Example: Terrain Data

Dataset: $\{(x_1, y_1, z_1), \dots, (x_k, y_k, z_k)\}$ where (x_i, y_i) is an longitude and latitude and z_i is an altitude

Find $\beta_0, \beta_1, \beta_2$ such that

$$f(x, y) = \beta_0 + \beta_1 x + \beta_2 y$$

which minimizes

$$\sum_{i=1}^k (f(x_i, y_i) - z_i)^2$$

$$\hat{\vec{\beta}} = (X^T X)^{-1} X^T \mathbf{z}$$

Step 3: Find the least squares solution of this system and use as the parameters of your model

An Aside: Unique Least Squares

$$\begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ \vdots & \vdots & \vdots \\ 1 & x_k & y_k \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_k \end{bmatrix}$$

A (Conceptual). Why can we almost always assume that the columns of this matrix are LI?

Answer

Answer

If the columns were linearly dependent, then one of our independent variables can be computed in terms of the others

Answer

If the columns were linearly dependent, then one of our independent variables can be computed in terms of the others

First off, this is very unlikely

Answer

If the columns were linearly dependent, then one of our independent variables can be computed in terms of the others

First off, this is very unlikely

Second, this variable could be then be thought of as a *dependent variable*

Answer

If the columns were linearly dependent, then one of our independent variables can be computed in terms of the others

First off, this is very unlikely

Second, this variable could be then be thought of as a *dependent* variable

It wouldn't contribute anything when using the least squares method

"Vectors" of Generalization



1. What if we have *more than one* independent value?

multiple regression, (hyper)plane of best fit

2. What if our data is not *exactly* linear.

e.g., polynomial regression

"Vectors" of Generalization



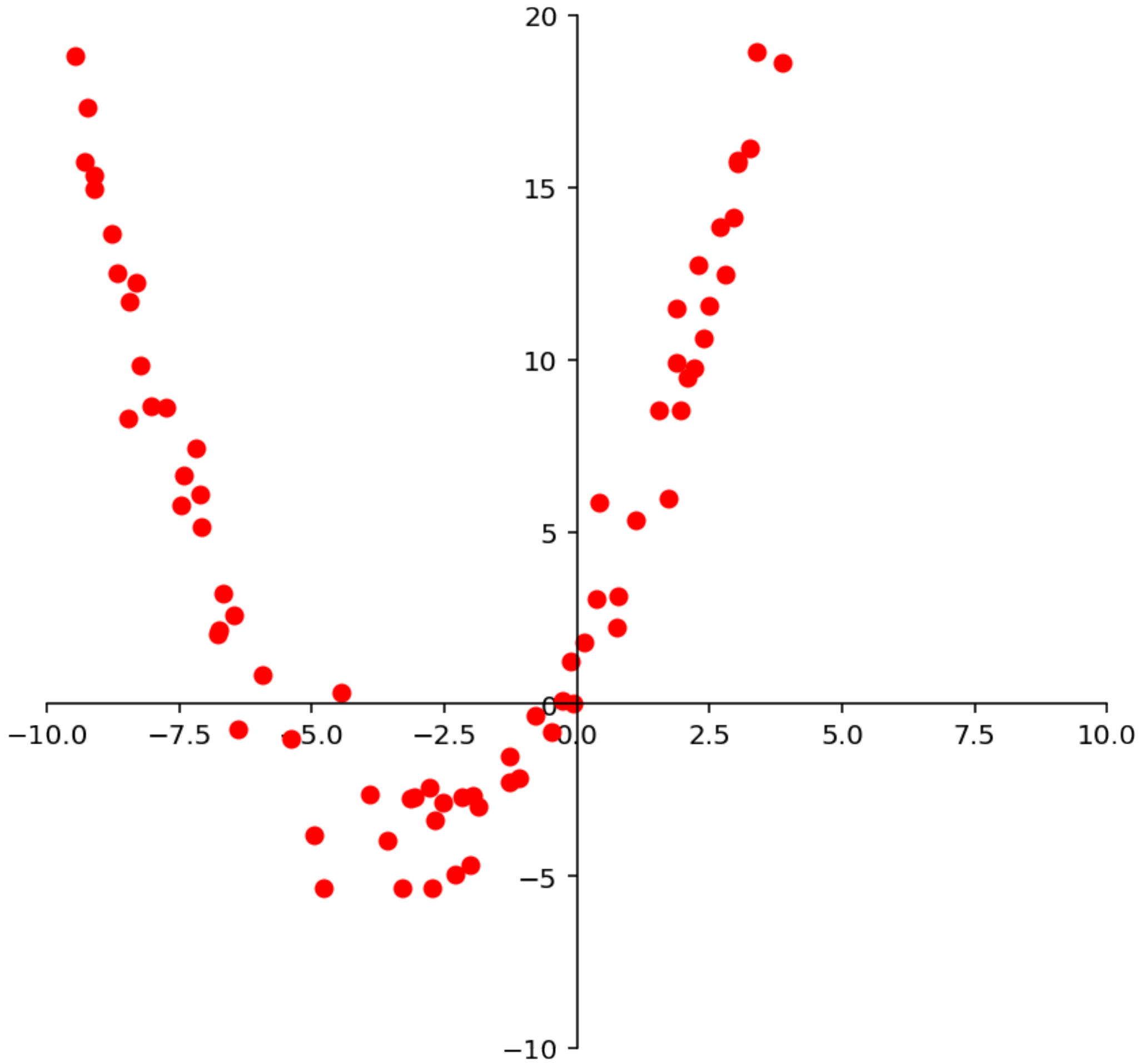
1. What if we have *more than one* independent value?

multiple regression, (hyper)plane of best fit

2. What if our data is not *exactly* linear.

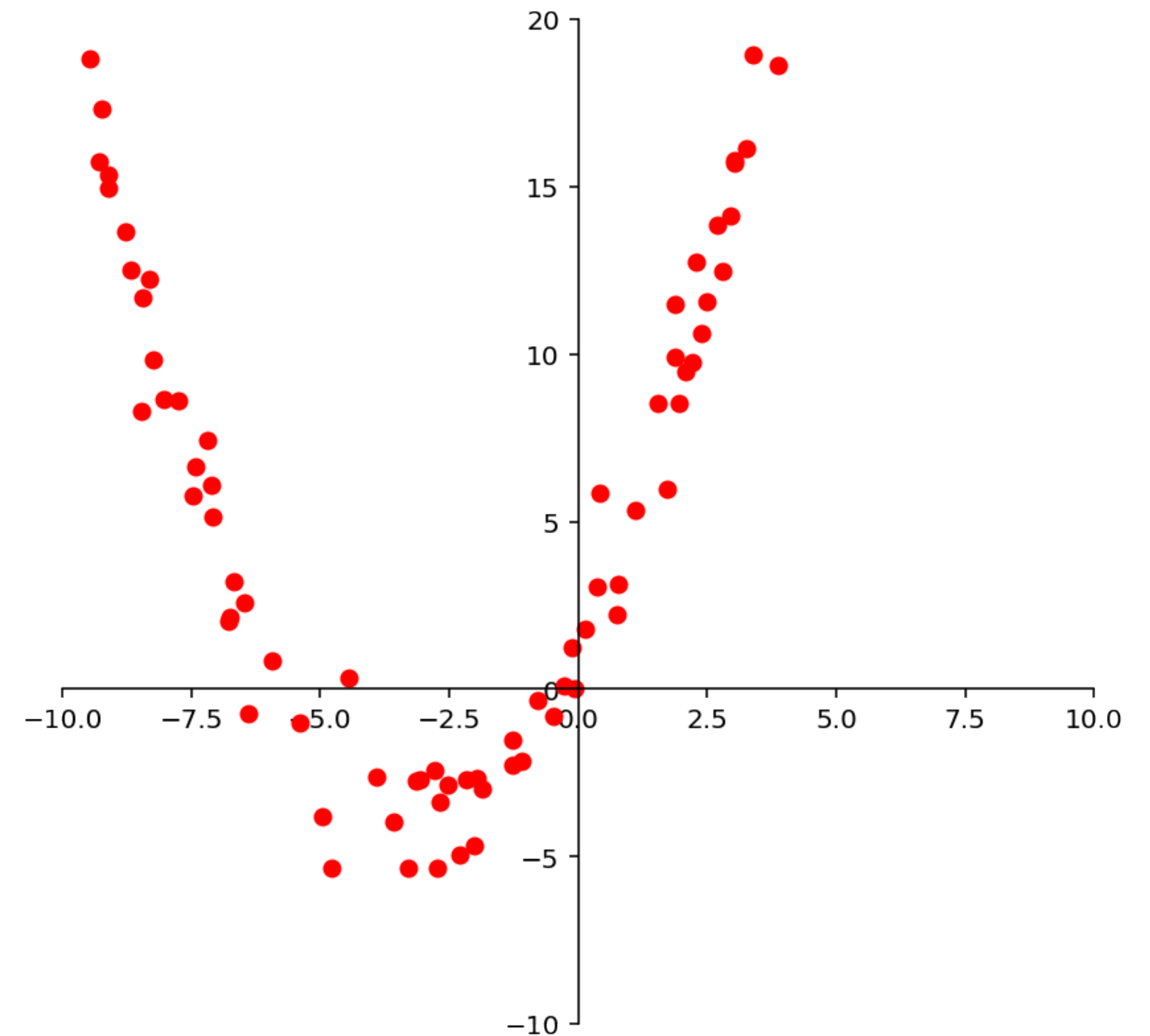
e.g., polynomial regression

Example: Best Fit Quadratic



Example: Best Fit Quadratic

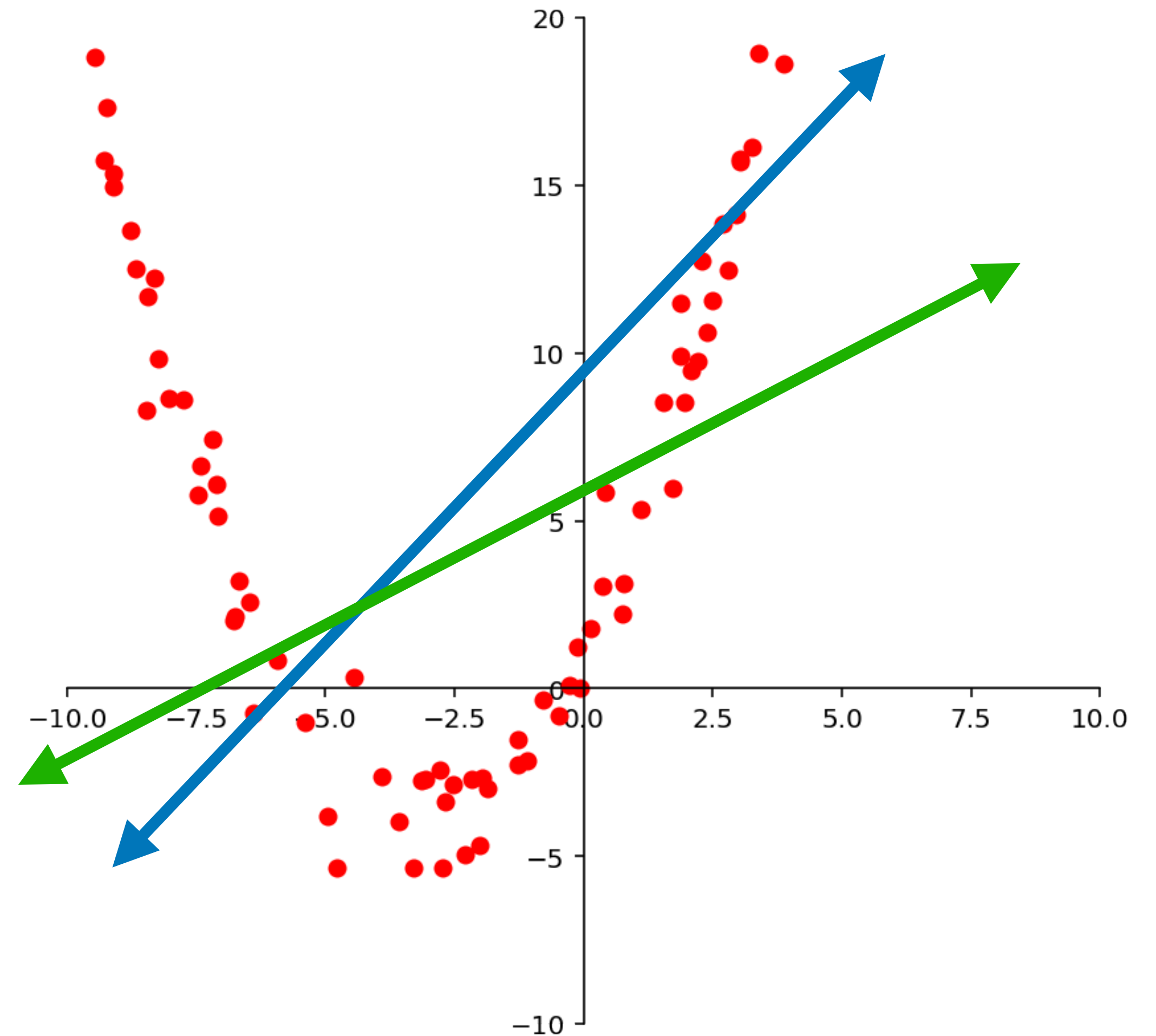
Dataset: $\{(x_1, y_1), \dots, (x_k, y_k)\}$



Example: Best Fit Quadratic

Dataset: $\{(x_1, y_1), \dots, (x_k, y_k)\}$

The issue: There is no good line to approximate this data

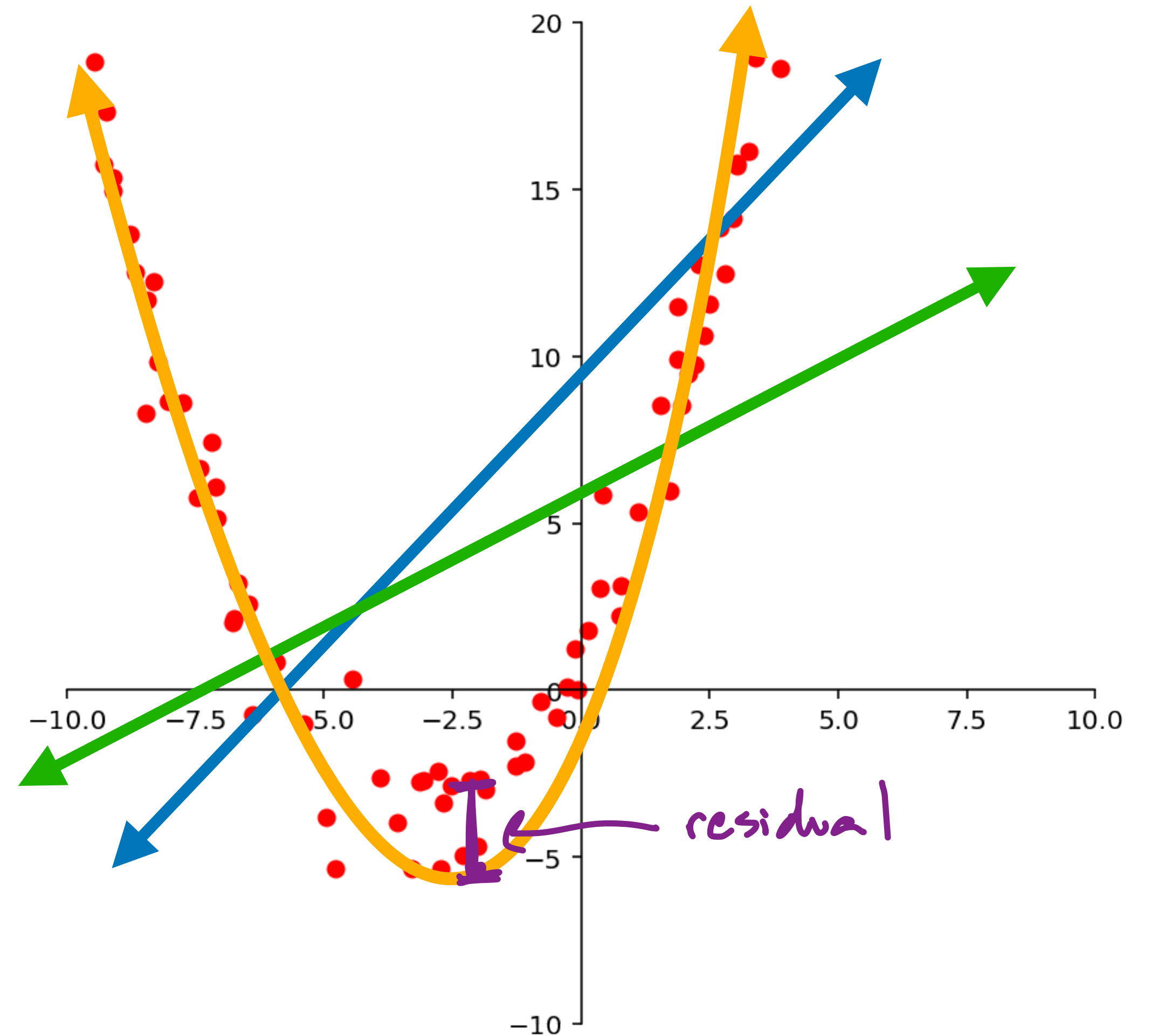


Example: Best Fit Quadratic

Dataset: $\{(x_1, y_1), \dots, (x_k, y_k)\}$

The issue: There is no good line to approximate this data

What about a parabola?



Example: Best Fit Quadratic

Dataset: $\{(x_1, y_1), \dots, (x_k, y_k)\}$

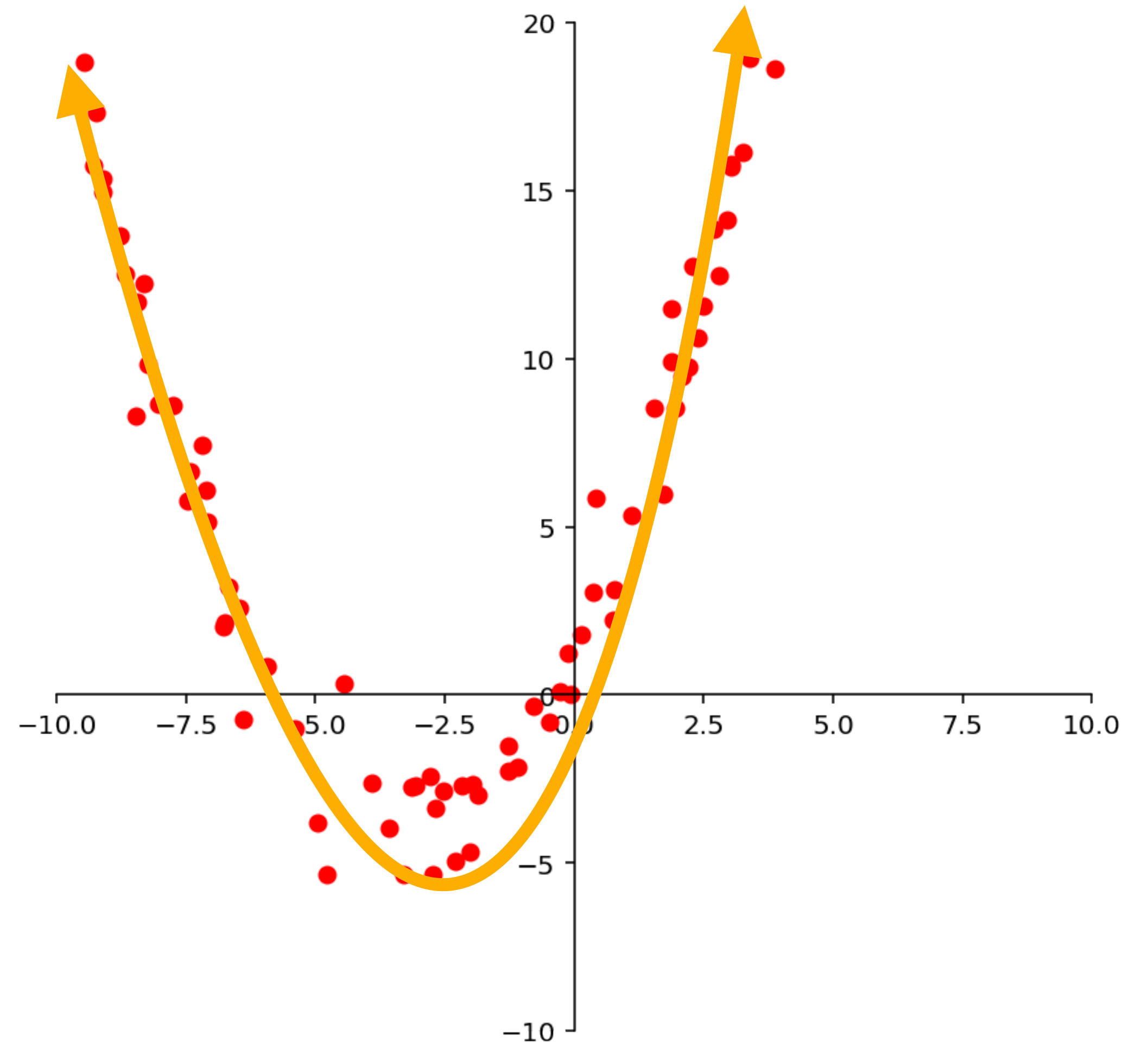
Find $\beta_0, \beta_1, \beta_2$ such that

$$f(x) = \boxed{\beta_0} + \boxed{\beta_1}x + \boxed{\beta_2}x^2$$

minimizes

parameters

$$\sum_{i=1}^k (f(x_i) - y_i)^2$$



Example: Best Fit Quadratic

Dataset: $\{(x_1, y_1), \dots, (x_k, y_k)\}$

Find $\beta_0, \beta_1, \beta_2$ such that

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2$$

minimizes

$$\sum_{i=1}^k (f(x_i) - y_i)^2$$

$$\beta_0 + \beta_1 x_1 + \beta_2 x_1^2 = y_1$$

$$\beta_0 + \beta_1 x_2 + \beta_2 x_2^2 = y_2$$

$$\vdots$$

$$\beta_0 + \beta_1 x_k + \beta_2 x_k^2 = y_k$$

Step 1: Set up an (almost assuredly inconsistent) system of linear equations in terms of the variables $\beta_0, \beta_1, \beta_2$

Example: Best Fit Quadratic

This is still linear in the β 's

Dataset: $\{(x_1, y_1), \dots, (x_k, y_k)\}$

Find $\beta_0, \beta_1, \beta_2$ such that

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2$$

minimizes

$$\sum_{i=1}^k (f(x_i) - y_i)^2$$

$$\beta_0 + \beta_1 x_1 + \beta_2 x_1^2 = y_1$$

$$\beta_0 + \beta_1 x_2 + \beta_2 x_2^2 = y_2$$

$$\boxed{1} \beta_0 + \boxed{x_2} \beta_1 + \boxed{x_2^2} \beta_2 = y_2$$

$$\beta_0 + 2\beta_1 + 4\beta_2 = 4$$
$$\beta_0 + \beta_1 x_k + \beta_2 x_k^2 = y_k$$

Step 1: Set up an (almost assuredly inconsistent) system of linear equations in terms of the variables $\beta_0, \beta_1, \beta_2$

Example: Best Fit Quadratic

Dataset: $\{(x_1, y_1), \dots, (x_k, y_k)\}$

Find $\beta_0, \beta_1, \beta_2$ such that

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2$$

minimizes

$$\sum_{i=1}^k (f(x_i) - y_i)^2$$

new column

$$\begin{bmatrix} 1 & \overset{\text{blue}}{X} x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_k & x_k^2 \end{bmatrix} \begin{bmatrix} \overset{\text{blue}}{\vec{\beta}} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} \overset{\text{blue}}{y} y_1 \\ y_2 \\ \vdots \\ y_k \end{bmatrix}$$

Step 2: Rewrite the system as a matrix equation

Example: Best Fit Quadratic

Dataset: $\{(x_1, y_1), \dots, (x_k, y_k)\}$

Find $\beta_0, \beta_1, \beta_2$ such that

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2$$

minimizes

$$\sum_{i=1}^k (f(x_i) - y_i)^2$$

$$\hat{\vec{\beta}} = (X^T X)^{-1} X^T \mathbf{y}$$

Step 3: Find the least squares solution of this system and use as the parameters of your model

The Takeaway

We can use non-linear modeling functions as long as they are *linear in the parameters*

Linear in Parameters

Def. A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is **linear in the parameters** $\beta_1, \dots, \beta_k$ if it can be written as

$$f(\mathbf{x}) = \beta_1 \phi_1(\mathbf{x}) + \beta_2 \phi_2(\mathbf{x}) + \dots + \beta_k \phi_k(\mathbf{x})$$

for functions $\phi_1, \dots, \phi_k: \mathbb{R}^n \rightarrow \mathbb{R}$

$$f(x, y) = \beta_1 \boxed{\sin(x^y)} + \beta_2 \boxed{\log x} - \beta_3 \boxed{1} + \beta_4 \boxed{x^x x}$$

We can build design matrices for
function which are linear in
their parameters

General Linear Regression

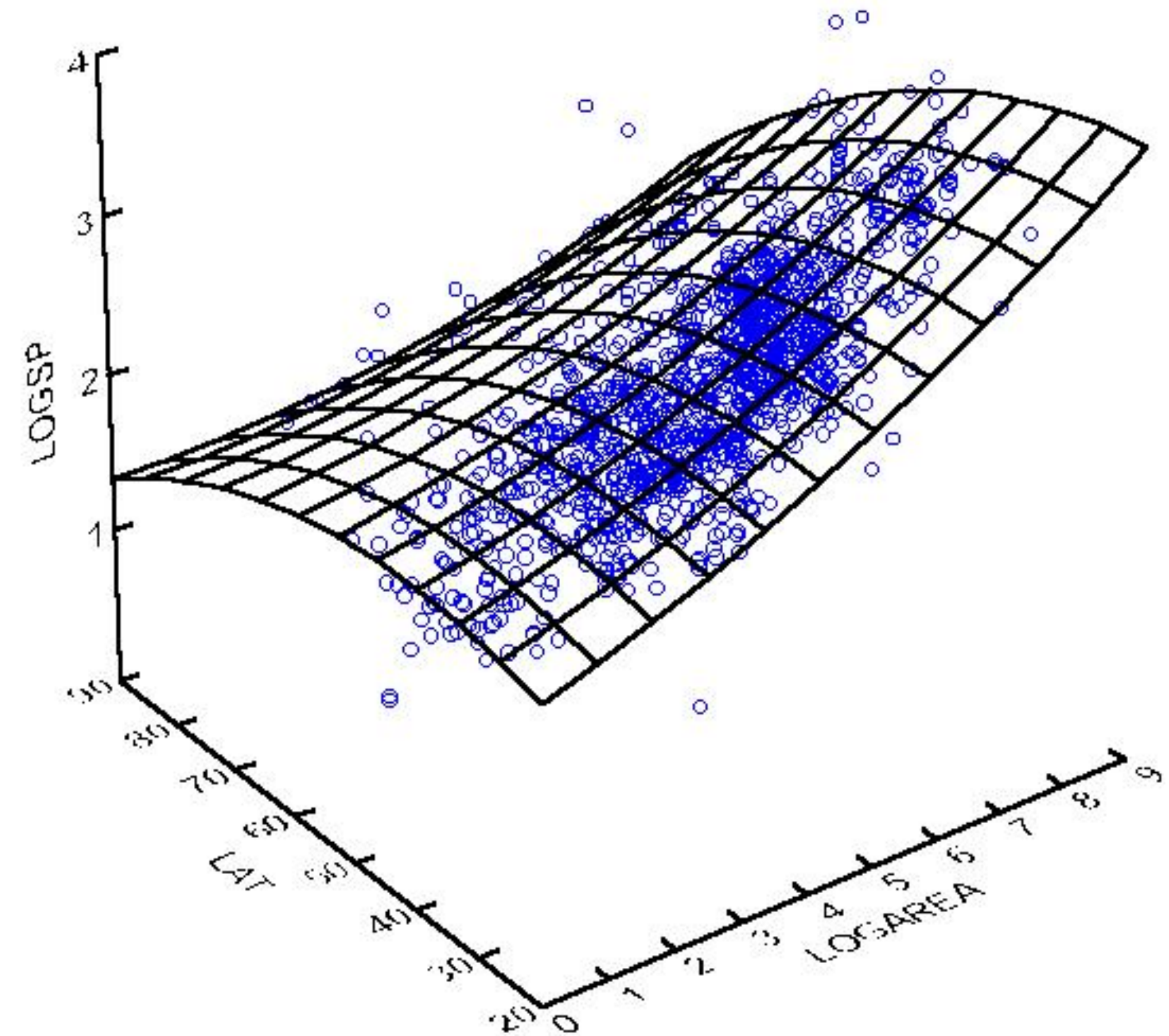
Dataset: $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m)\}$ where $\mathbf{x}_i \in \mathbb{R}^n$ and $y_i \in \mathbb{R}$

Given a function

$$f_{\beta_1, \dots, \beta_k} : \mathbb{R}^n \rightarrow \mathbb{R}$$

which is *linear in the parameters* $\beta_1, \dots, \beta_k$, find values for $\beta_1, \dots, \beta_k$ which minimize

$$\sum_{i=1}^k (f_{\beta_1, \dots, \beta_k}(\mathbf{x}_i) - y_i)^2$$



General Linear Regression

Dataset: $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m)\}$ where $\mathbf{x}_i \in \mathbb{R}^n$ and $y_i \in \mathbb{R}$

Given a function

$$f_{\beta_1, \dots, \beta_k} : \mathbb{R}^n \rightarrow \mathbb{R}$$

which is *linear in the parameters* $\beta_1, \dots, \beta_k$, find values for $\beta_1, \dots, \beta_k$ which minimize

$$\sum_{i=1}^k (f_{\beta_1, \dots, \beta_k}(\mathbf{x}_i) - y_i)^2$$

$$\begin{aligned}\beta_1 \phi_1(\mathbf{x}_1) + \dots + \beta_k \phi_k(\mathbf{x}_1) &= y_1 \\ \beta_1 \phi_1(\mathbf{x}_2) + \dots + \beta_k \phi_k(\mathbf{x}_2) &= y_2 \\ &\vdots \\ \beta_1 \phi_1(\mathbf{x}_n) + \dots + \beta_k \phi_k(\mathbf{x}_n) &= y_n\end{aligned}$$

Step 1: Set up an (almost assuredly inconsistent) system of linear equations in terms of the variables $\beta_1, \dots, \beta_k$

General Linear Regression

This is still linear in the β 's

Dataset: $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m)\}$ where $\mathbf{x}_i \in \mathbb{R}^n$ and $y_i \in \mathbb{R}$

Given a function

$$f_{\beta_1, \dots, \beta_k} : \mathbb{R}^n \rightarrow \mathbb{R}$$

which is *linear in the parameters* $\beta_1, \dots, \beta_k$, find values for $\beta_1, \dots, \beta_k$ which minimize

$$\sum_{i=1}^k (f_{\beta_1, \dots, \beta_k}(\mathbf{x}_i) - y_i)^2$$

$$\begin{aligned}\beta_1 \phi_1(\mathbf{x}_1) + \dots + \beta_k \phi_k(\mathbf{x}_1) &= y_1 \\ \beta_1 \phi_1(\mathbf{x}_2) + \dots + \beta_k \phi_k(\mathbf{x}_2) &= y_2 \\ &\vdots \\ \beta_1 \phi_1(\mathbf{x}_m) + \dots + \beta_k \phi_k(\mathbf{x}_m) &= y_m\end{aligned}$$

Step 1: Set up an (almost assuredly inconsistent) system of linear equations in terms of the variables $\beta_1, \dots, \beta_k$

General Linear Regression

Dataset: $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m)\}$ where $\mathbf{x}_i \in \mathbb{R}^n$
and $y_i \in \mathbb{R}$

Given a function

$$f_{\beta_1, \dots, \beta_k} : \mathbb{R}^n \rightarrow \mathbb{R}$$

which is *linear in the parameters*
 $\beta_1, \dots, \beta_k$, find values for $\beta_1, \dots, \beta_k$ which
minimize

$$\sum_{i=1}^k (f_{\beta_1, \dots, \beta_k}(\mathbf{x}_i) - y_i)^2$$

design matrix

$$\overset{X}{\begin{bmatrix} \phi_1(\mathbf{x}_1) & \phi_2(\mathbf{x}_1) & \dots & \phi_k(\mathbf{x}_1) \\ \phi_1(\mathbf{x}_2) & \phi_2(\mathbf{x}_2) & \dots & \phi_k(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(\mathbf{x}_m) & \phi_2(\mathbf{x}_m) & \dots & \phi_k(\mathbf{x}_m) \end{bmatrix}} \begin{bmatrix} \vec{\beta} \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{bmatrix} = \begin{bmatrix} \mathbf{y} \\ y_1 \\ y_2 \\ \vdots \\ y_k \end{bmatrix}$$

Step 2: Rewrite the system as a
matrix equation

General Linear Regression

Dataset: $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m)\}$ where $\mathbf{x}_i \in \mathbb{R}^n$
and $y_i \in \mathbb{R}$

Given a function

$$f_{\beta_1, \dots, \beta_k} : \mathbb{R}^n \rightarrow \mathbb{R}$$

which is *linear in the parameters*
 $\beta_1, \dots, \beta_k$, find values for $\beta_1, \dots, \beta_k$ which
minimize

$$\sum_{i=1}^k (f_{\beta_1, \dots, \beta_k}(\mathbf{x}_i) - y_i)^2$$

$$\hat{\vec{\beta}} = (X^T X)^{-1} X^T \mathbf{y}$$

Step 3: Find the least squares
solution of this system and use
as the parameters of your model

How To: Design Matrices

How To: Design Matrices

Q. Find the design matrix for least squares regression with the function f in terms of the parameters $\beta_1, \beta_2, \dots, \beta_k$ given the dataset $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m)\}$

How To: Design Matrices

Q. Find the design matrix for least squares regression with the function f in terms of the parameters $\beta_1, \beta_2, \dots, \beta_k$ given the dataset $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m)\}$

A. First write $f(\mathbf{x})$ as $\beta_1\phi_1(\mathbf{x}) + \dots + \beta_k\phi_k(\mathbf{x})$ where $\phi_1, \dots, \phi_k$ are potentially non-linear functions. Then build the matrix:

$$\begin{bmatrix} \phi_1(\mathbf{x}_1) & \phi_2(\mathbf{x}_1) & \dots & \phi_k(\mathbf{x}_1) \\ \phi_1(\mathbf{x}_2) & \phi_2(\mathbf{x}_2) & \dots & \phi_k(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(\mathbf{x}_m) & \phi_2(\mathbf{x}_m) & \dots & \phi_k(\mathbf{x}_m) \end{bmatrix}$$

Example

$$f(x_1, x_2, x_3) = \beta_1 (\overset{\phi_1(x_1, x_2, x_3)}{\cos(x_1)} - x_3) + \beta_2 (\overset{\phi_2(x_1, x_2, x_3)}{e^{-x_1 x_2}}) + \beta_3 (\overset{\phi_3(x_1, x_2, x_3)}{1})$$

Find the design matrix for the least squares regression with the function

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \beta_1 \cos(x_1) + \beta_2 e^{-x_1 x_2} - \beta_1 x_3 + \beta_3$$

for the dataset

$$\cos \pi = -1$$

$$\begin{bmatrix} 1 & 1 & 1 \\ -1 & e^{-3\pi} & 1 \end{bmatrix} = \begin{bmatrix} \phi_1(0,0,0) & \phi_2(0,0,0) & \phi_3(0,0,0) \\ \phi_1(\pi,3,1) & \phi_2(\pi,3,1) & \phi_3(\pi,3,1) \end{bmatrix}$$

$\mathbf{x}_1 = (0,0,0) \quad y_1 = 5$
 $\mathbf{x}_2 = (\pi,3,1) \quad y_2 = 3$

We will see more of this
in Lab 6

Practical Considerations

Practical Considerations

Many functions require large design matrices, e.g.
multivariate polynomials have *a lot* of possible terms

Practical Considerations

Many functions require large design matrices, e.g.
multivariate polynomials have *a lot* of possible terms

We haven't actually talked about *which* modeling
functions to use

Practical Considerations

Many functions require large design matrices, e.g. multivariate polynomials have *a lot* of possible terms

We haven't actually talked about *which* modeling functions to use

Again, is least-squares error really what we want? What if we want to minimize something else?

Practical Considerations

Many functions require large design matrices, e.g. multivariate polynomials have *a lot* of possible terms

We haven't actually talked about *which* modeling functions to use

Again, is least-squares error really what we want? What if we want to minimize something else?

Concerns for another class